Bioinformatics Institute
A*STAR BII

AI/ML in Pharmacometrics

25 October 2026

James Lu

A*STAR Bioinformatics Institute

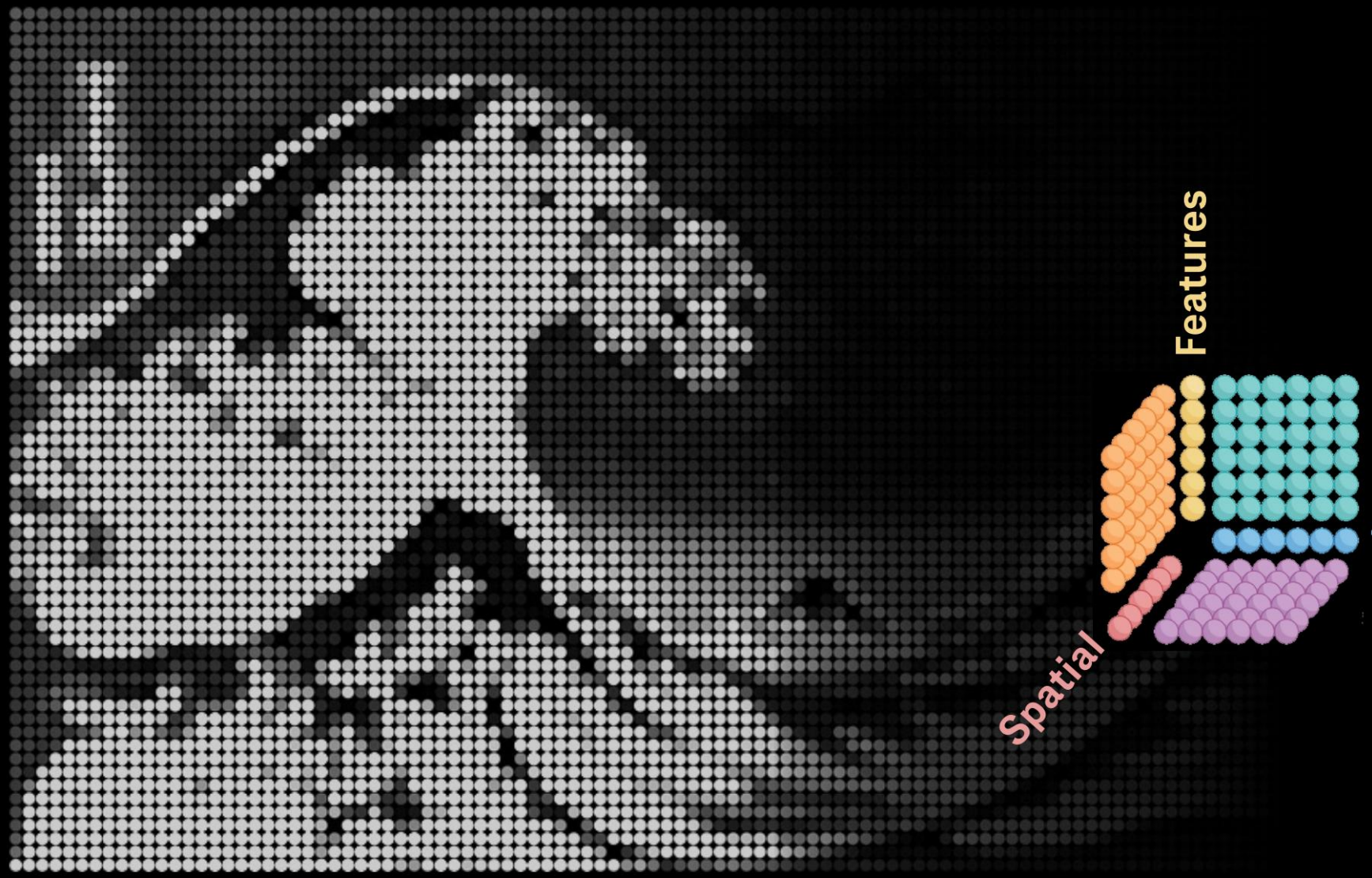


Agenda



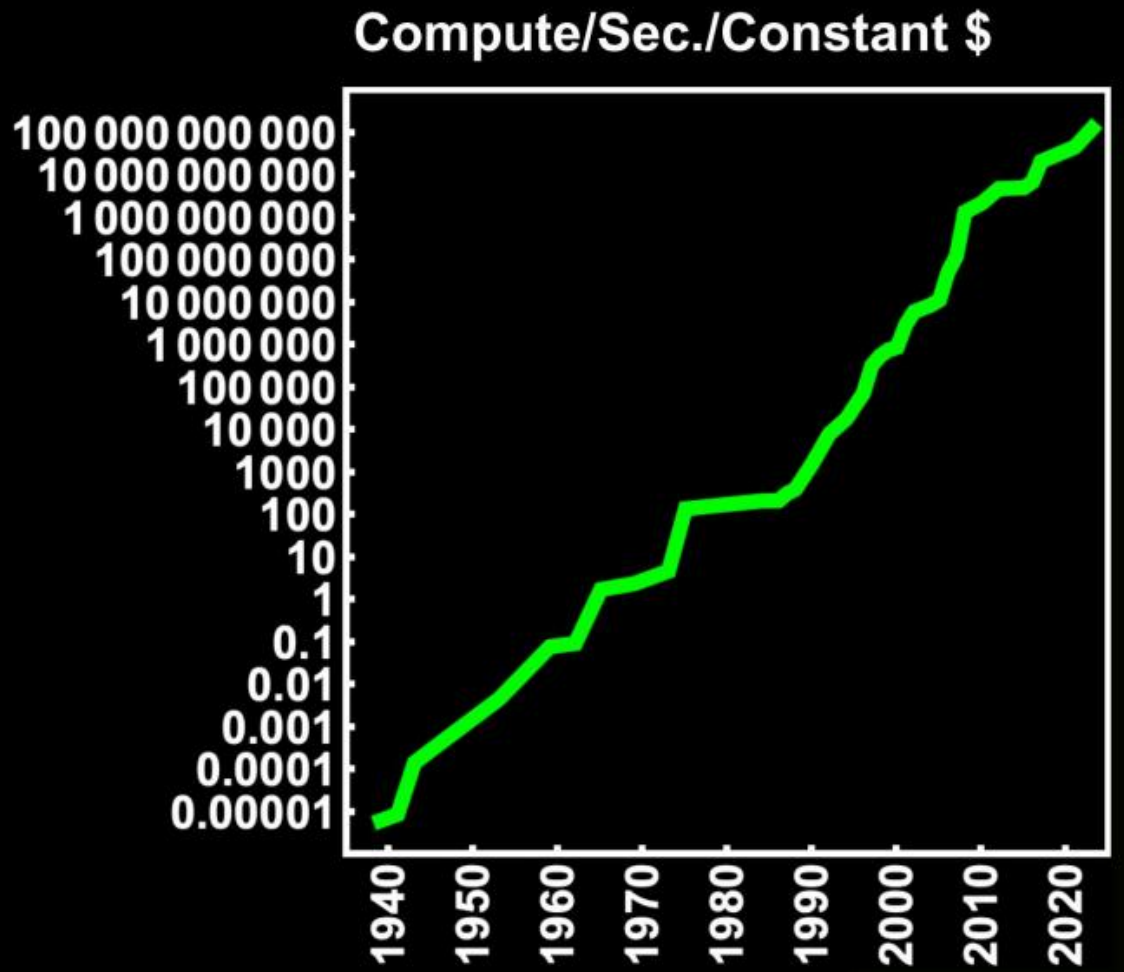
Time	
9:00-9:45 AM	Lecture 1
9:45-10:00 AM	Break
10:00-10:30 AM	Hands-On
10:30-11:15 AM	Lecture 2
11:15-11:30 AM	Break
11:30 AM-12:00 PM	Outlook/ Discussions

The Oncoming Wave of Multimodal Biomedical Data

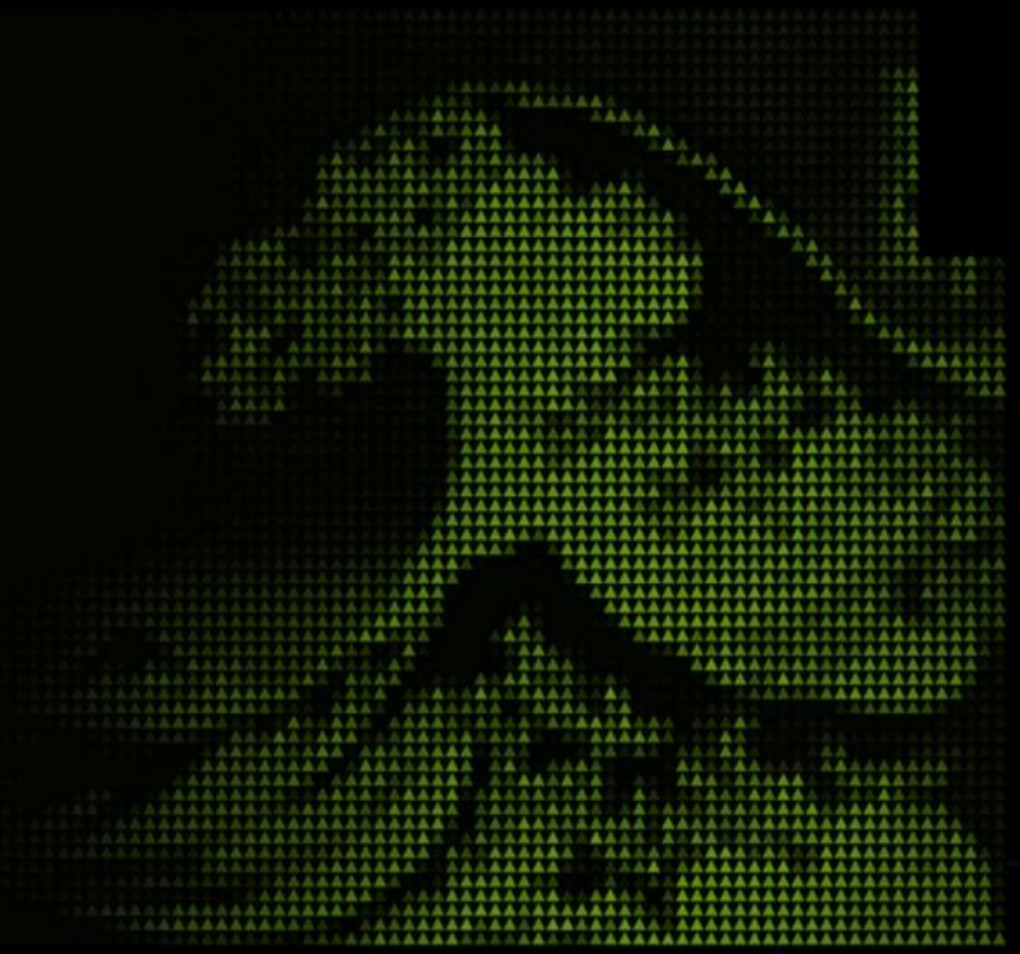


e.g.-
RWD
Digital Health
Imaging
RNA-seq
Single Cell Omics
Spatial Omics
Organoids
…

The Oncoming Wave of Compute

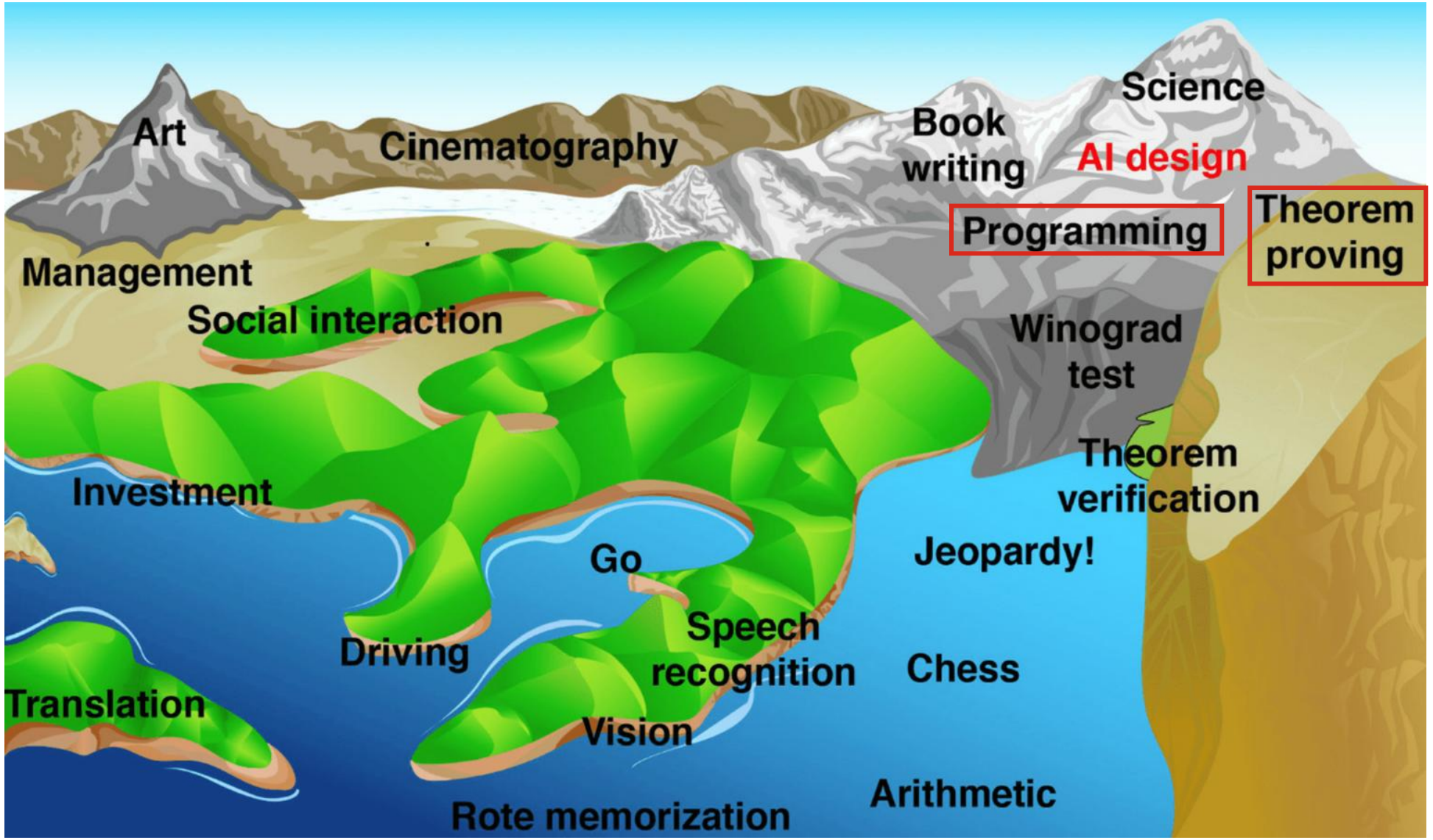


Ray Kurzweil: The Singularity is Nearer (2024)



The Landscape of Human Competence

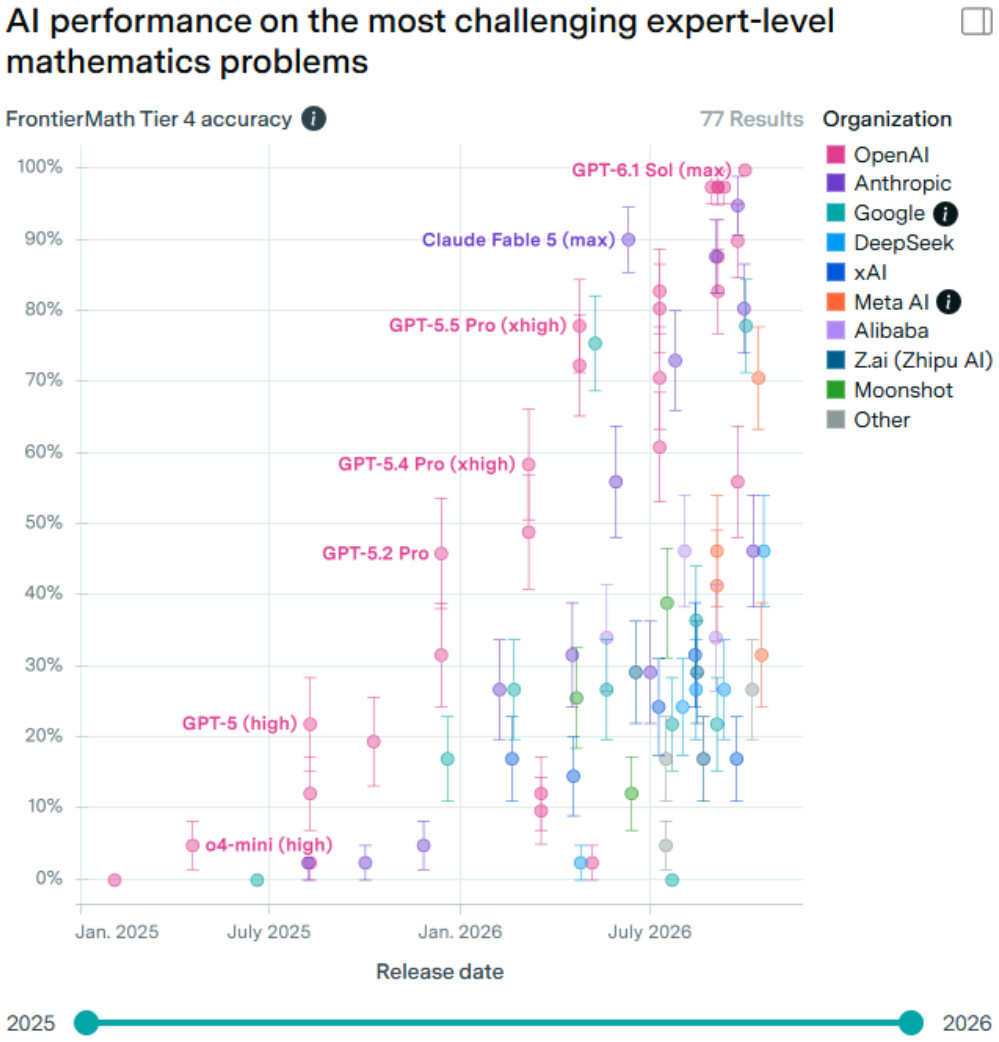
Life 3.0 – Max Tegmark (2017)



Where has AI capability risen to?

AI in Theorem Proving

Increase in capability over a short time …



QUALIA

Is AI the End of Math As We Know It?

Mathematicians are facing the sudden shift with grief, anger, and a desperate search for fresh ideas: "If we don't adapt, there's just no more math in 50 years."

31 | | Add as Preferred Source



Society for Mathematical Biology

SMB 2027 Annual Meeting

27 June 2027 - 2 July 2027 | San Diego, CA



What will be the most pressing topic for Scientific Programming of SMB?

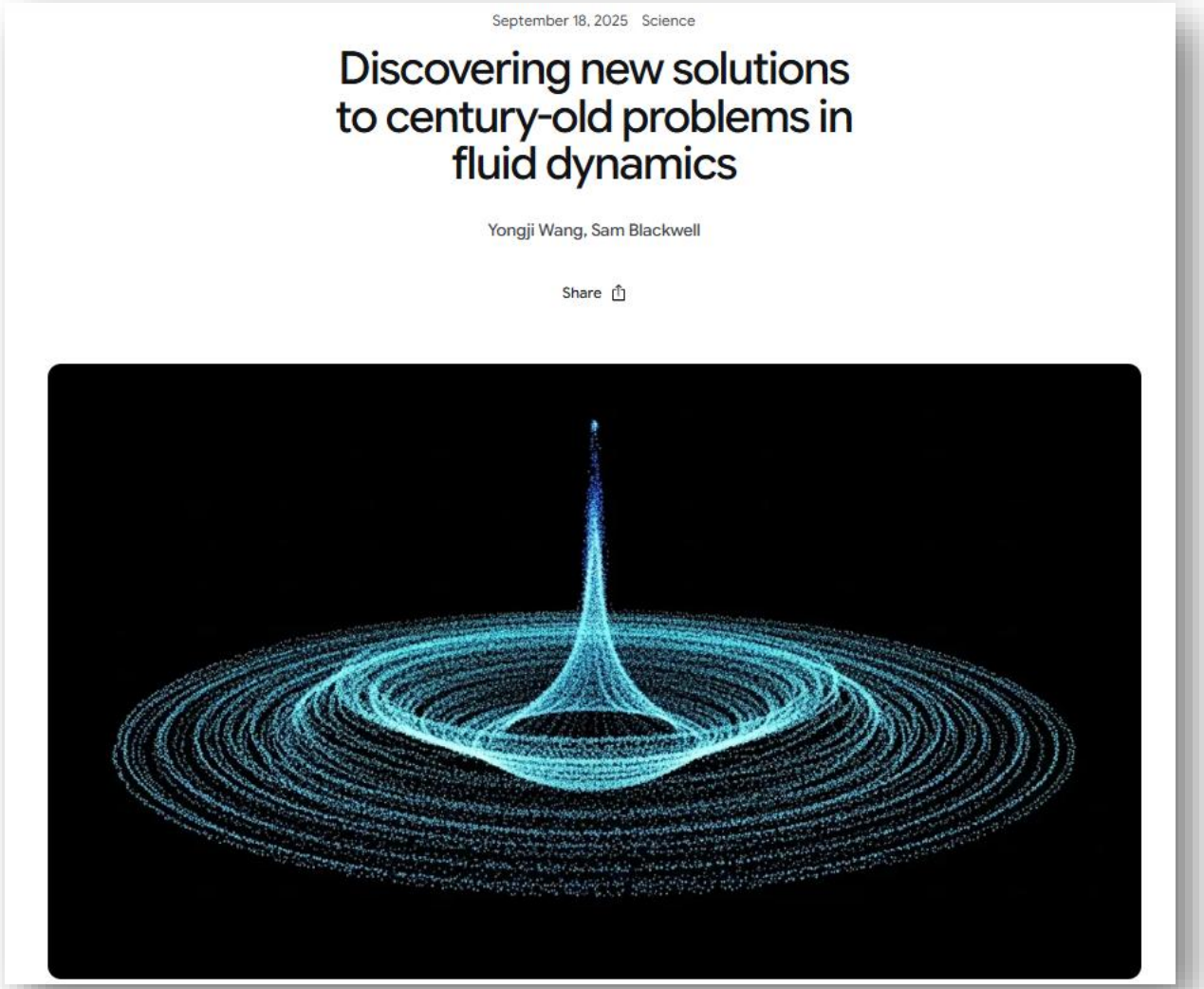
AI in Theorem Proving

Discuss: what are the likely impacts on PMx & QSP?

Era	AI Math Milestone	Impact on Pharmacometrics & QSP
Early 2024	**Pattern Matching:** Struggled with multi-step competition math and basic calculus derivations.	**?**
2025	**Test-Time Reasoning:** Hit **~90% on AIME** and **~40% on IMO proofs** via reinforcement learning.	**?**
2026	**Research & Formal Proofs:** Solving live **ArXivMath** papers and verifying formal theorems in Lean 4.	**?**

AI in Theorem Proving

Can (Agentic) AI use novel AI/ML methods to make new discoveries?

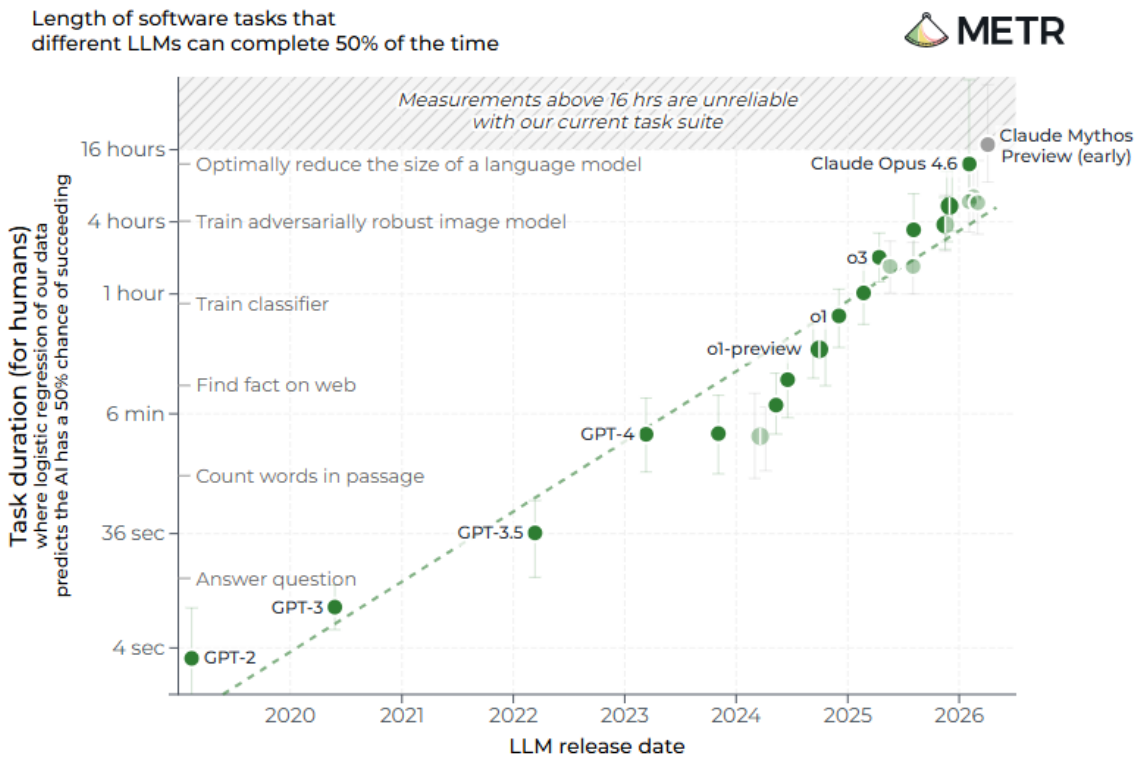


“By embedding mathematical insights and achieving extreme precision, we transformed PINNs into a discovery tool that finds elusive singularities.”

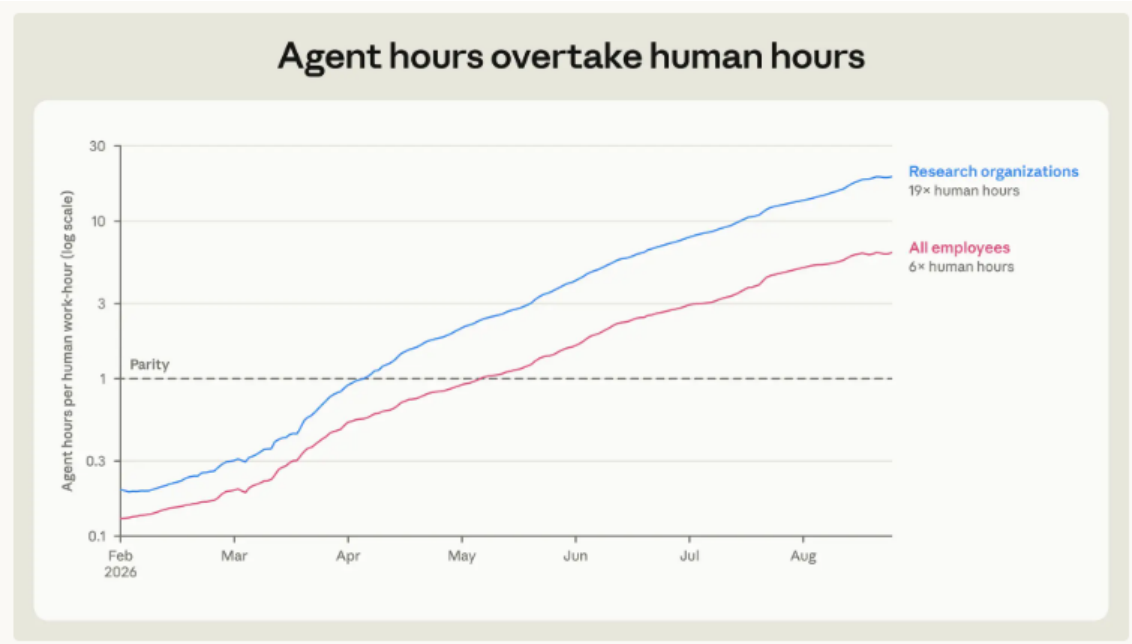
Yongji Wang
first author of the study and Postdoctoral Researcher at NYU

AI in Coding

Increase in capability over a short time …



Anthropic



Andrej Karpathy: Software 3.0, Vibe Coding, and the Future of Engineering

The Shift to Software 3.0

Computing paradigms are evolving rapidly:

Software 1.0: Traditional programming relying on explicit, hand-written rules.
Software 2.0: The deep learning era, where neural networks learn from data via weights.
Software 3.0: The agentic era, where prompting an LLM acts as the core programming

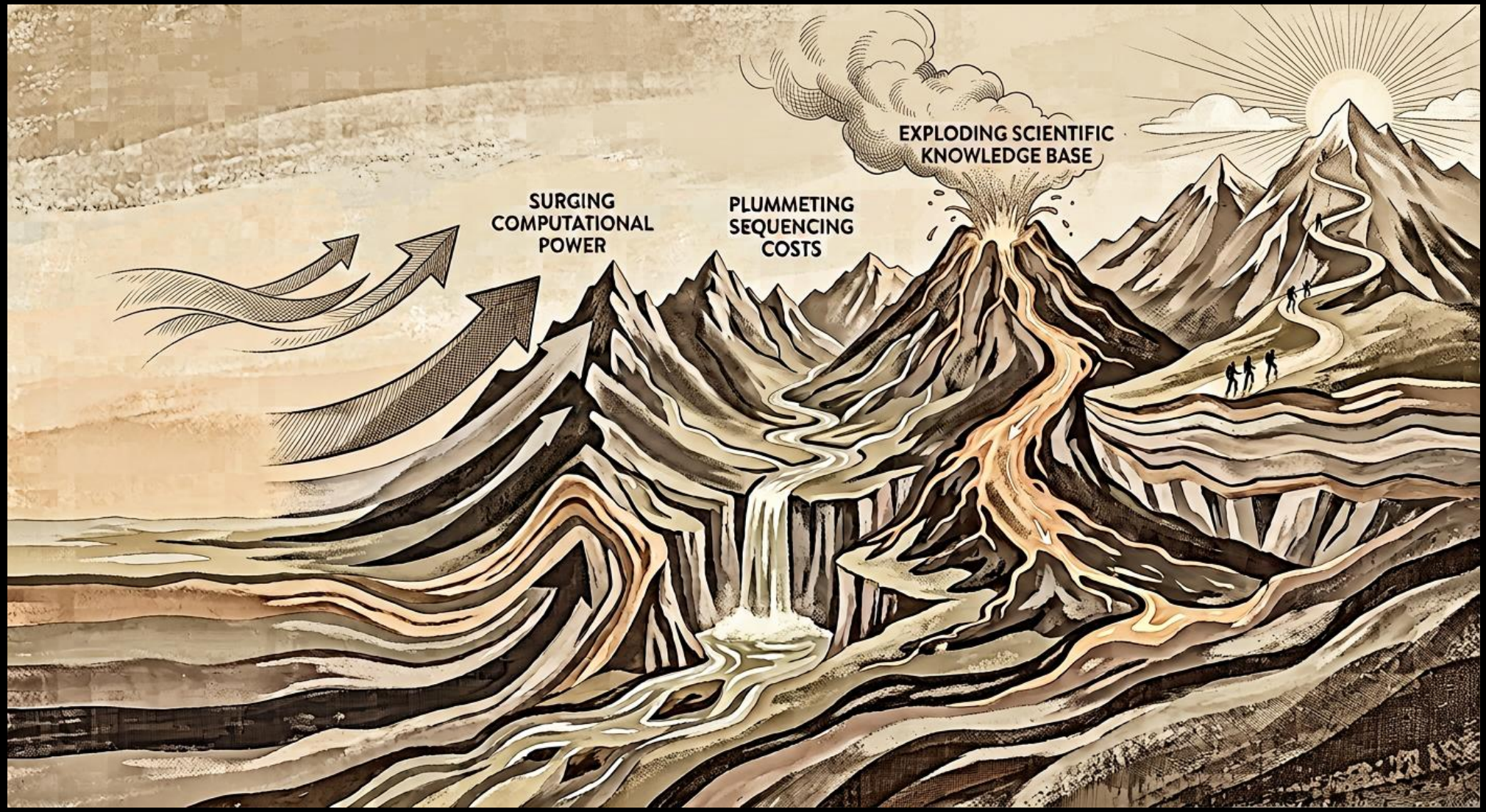
AI in Coding

Discuss: what are the likely impacts on PMx & QSP?

Era	AI Coding Milestone	Benchmark Reality (`SWE-bench Verified`)	Impact on Pharmacometrics & QSP
Early 2024	**Autocomplete & Snippets:** Context limited to single scripts; hallucinated package APIs and could not run or test its own code.	**~4% – 15% resolved** *(Failed on multi-file repositories)*	Useful for writing boilerplate `ggplot2` code, regex data cleaning, or basic R-to-NONMEM syntax translation.
2025	**Execution Loops & CLI Agents:** Reasoning models coupled with terminal access to read tracebacks, run unit tests, and iteratively fix bugs.	**~50% – 65% resolved** *(Handled standard repository issues)*	Automated CDISC/SDTM dataset assembly, custom PyTorch/Julia ODE solver scripts, and automated diagnostic plotting.
2026	**Multi-Agent Repo Orchestration:** Parallel agent swarms architecting full codebases, running long-horizon simulations, and enforcing formal verification.	**80%+ resolved** *(Saturated; shifted to `SWE-bench Pro` & `SciCode`)*	**?**

The Changing Landscape of Drug Discovery & Development

The convergence of exponential technological forces



Levels of Human-Machine Partnership

Machine as **Partner**

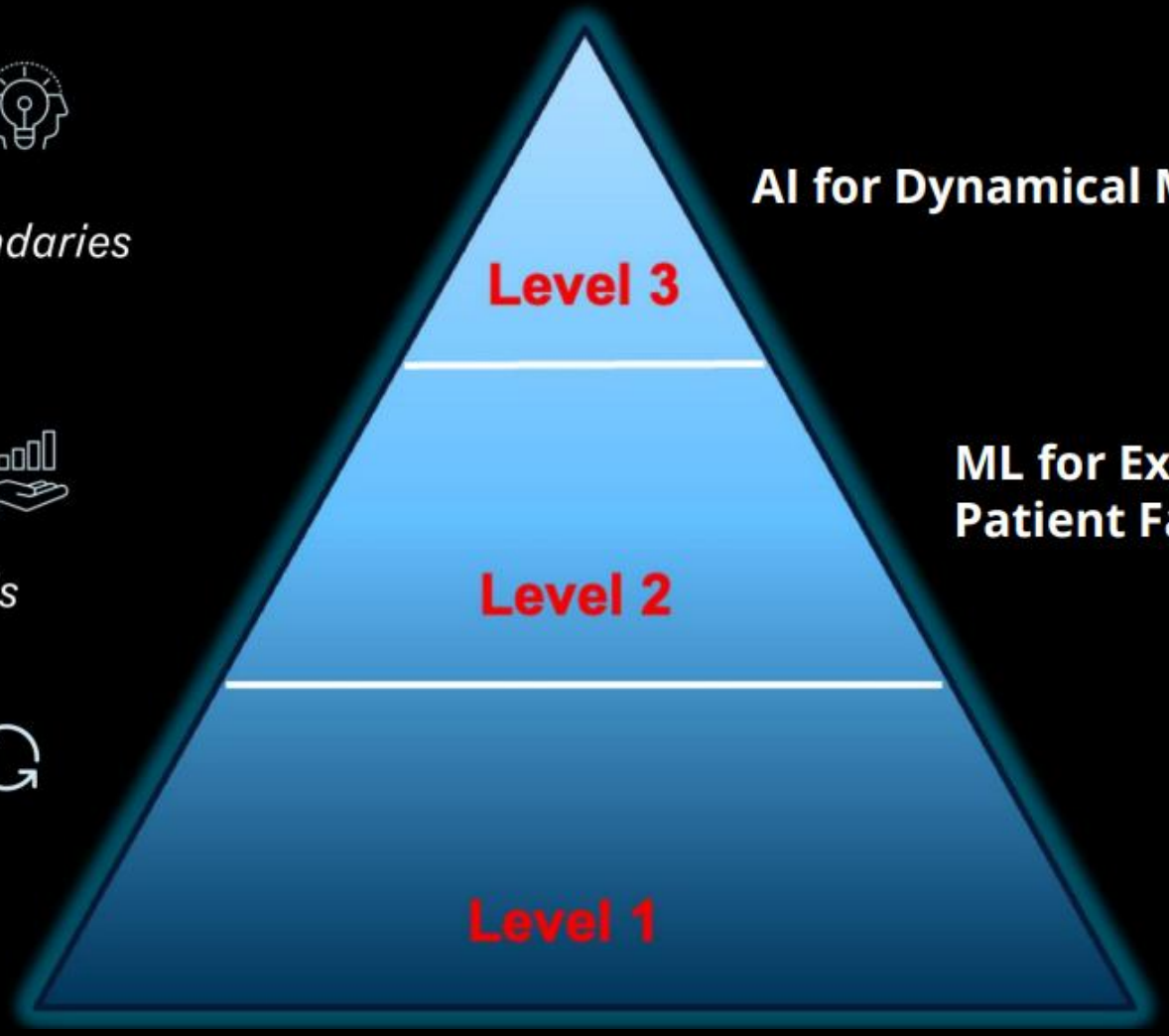
Push the boundaries

Machine as **Assistant**

Improve models

Machine as **Tool**

Save-Time



AI for Dynamical Model Discovery

ML for Exposure-Response & Patient Factor Identification

Artificial Intelligence for Quantitative Modeling in Drug Discovery and Development: An Innovation and Quality Consortium Perspective on Use Cases and Best Practices: CPT (2023)

What AI developments can leverage the growth in data?

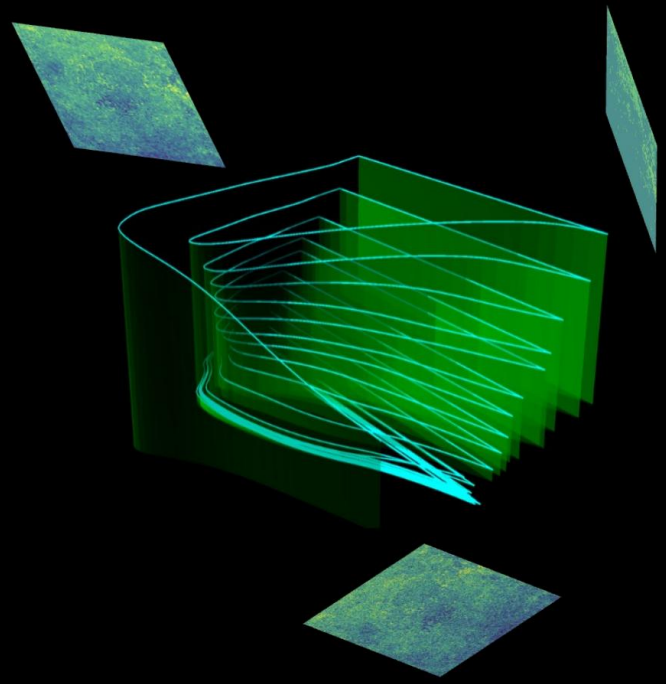
Extend Language

$$\frac{dy(t)}{dt} = \left(\text{NN}\right)(y(t), p)$$



e.g., Neural Differential Equations

Generalize Concepts



e.g., Learned Vector Embedding

Augment Intelligence

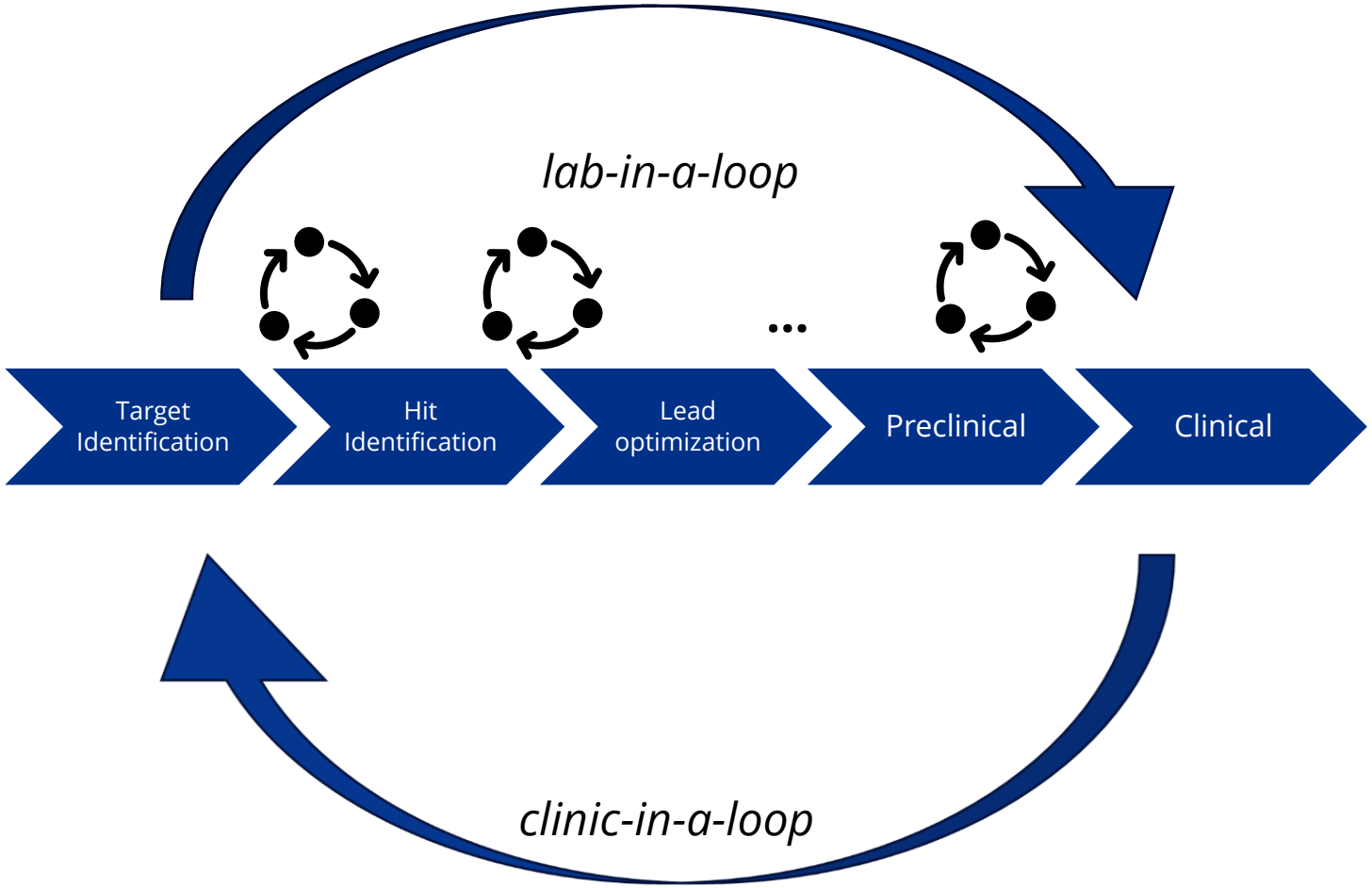


e.g., AI Agents

Opportunities for multi-level cycles of predictions & learning

... enabled by AI

Model-Informed Drug Discovery & Development



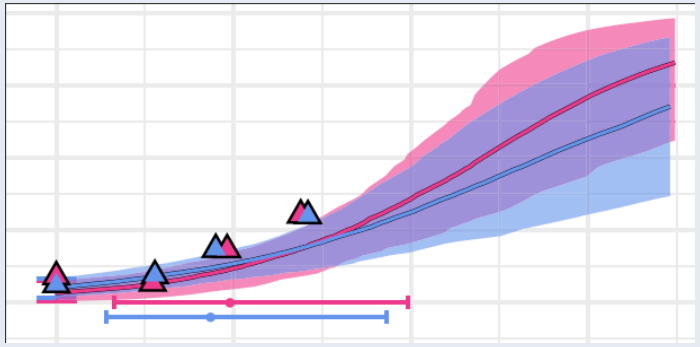
Reverse Translation

The Two Worlds of E-R Analysis

Pharmacometrics

- Typically trial-by-trial basis
- E-R relationship selected from a few pre-defined functions
- A handful of covariates are pre-selected

*e.g., **3** covariates, assuming **linear** exposure*

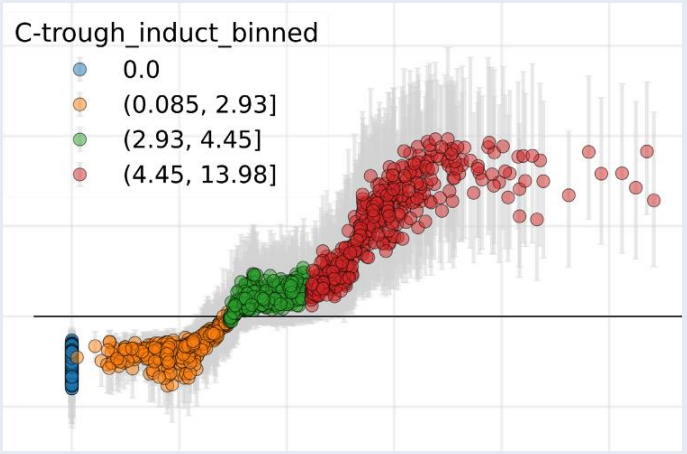


Kassir *et al*. CPT:PSP (2022)

Machine Learning

- Often examined across trials
- E-R relationship “learned” from data
- Covariates are typically comprehensively utilized

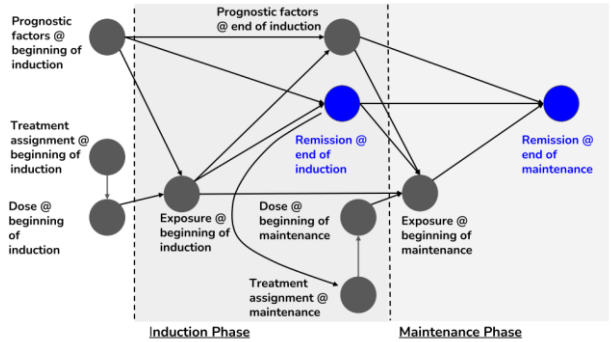
*e.g., **68** covariates and exposure*



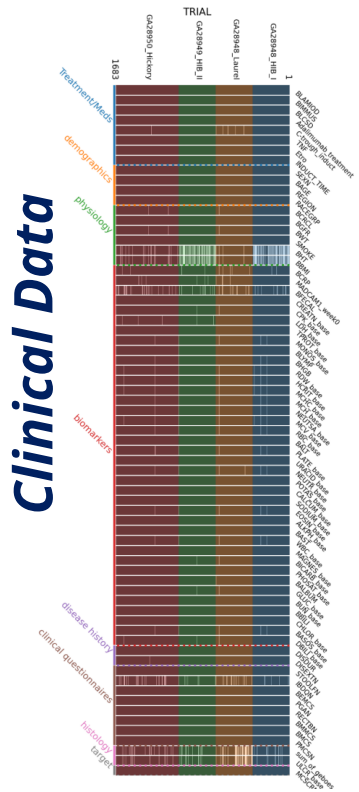
Harun *et al*. CPT:PSP (2024)

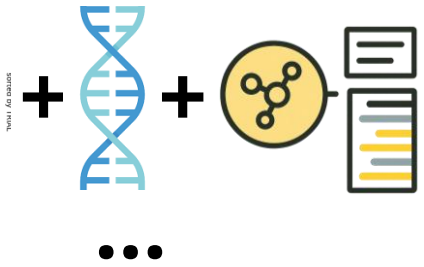
Identifying Patient Factors & Exposure-Response From Clinical Data

Causal Diagram

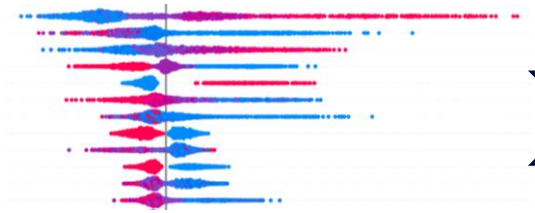


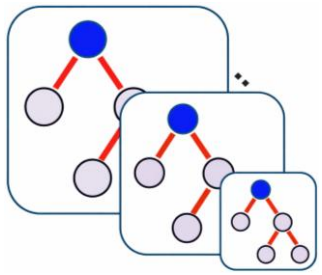
Clinical Data



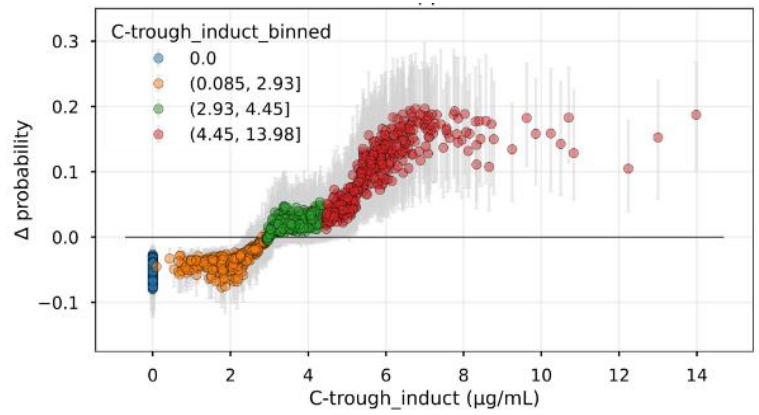


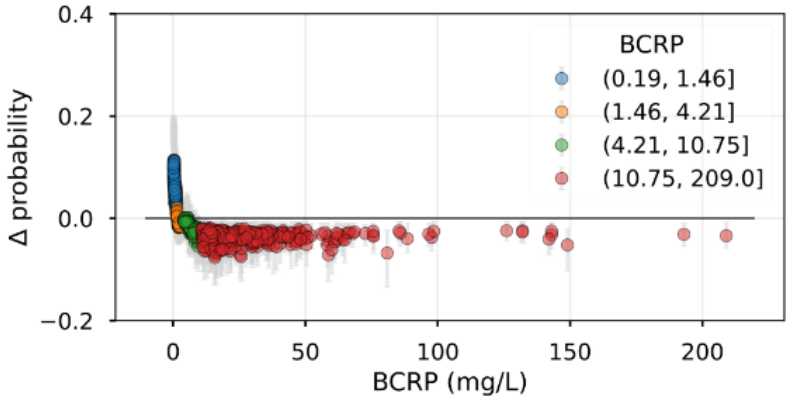
Causally aware ML model





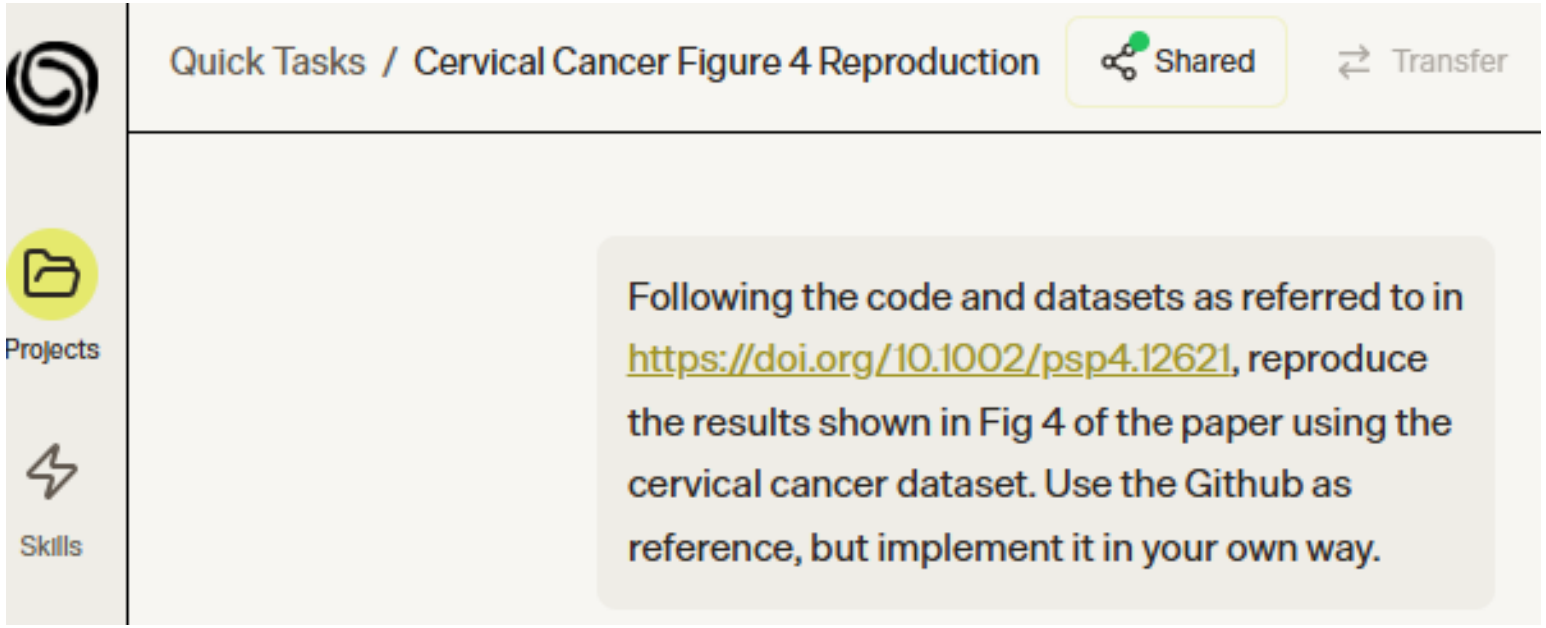
Learned relationships Efficacy & Safety





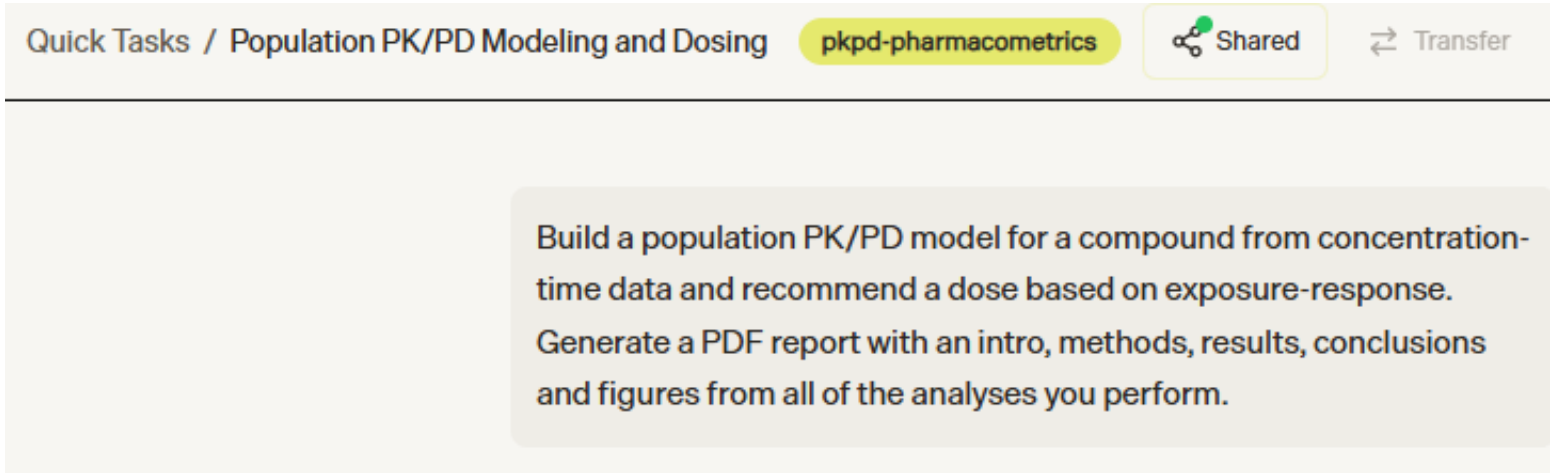
Harun R *et al*. Clinical Pharmacology & Therapeutics (2024)

Biomni example: Classification Task

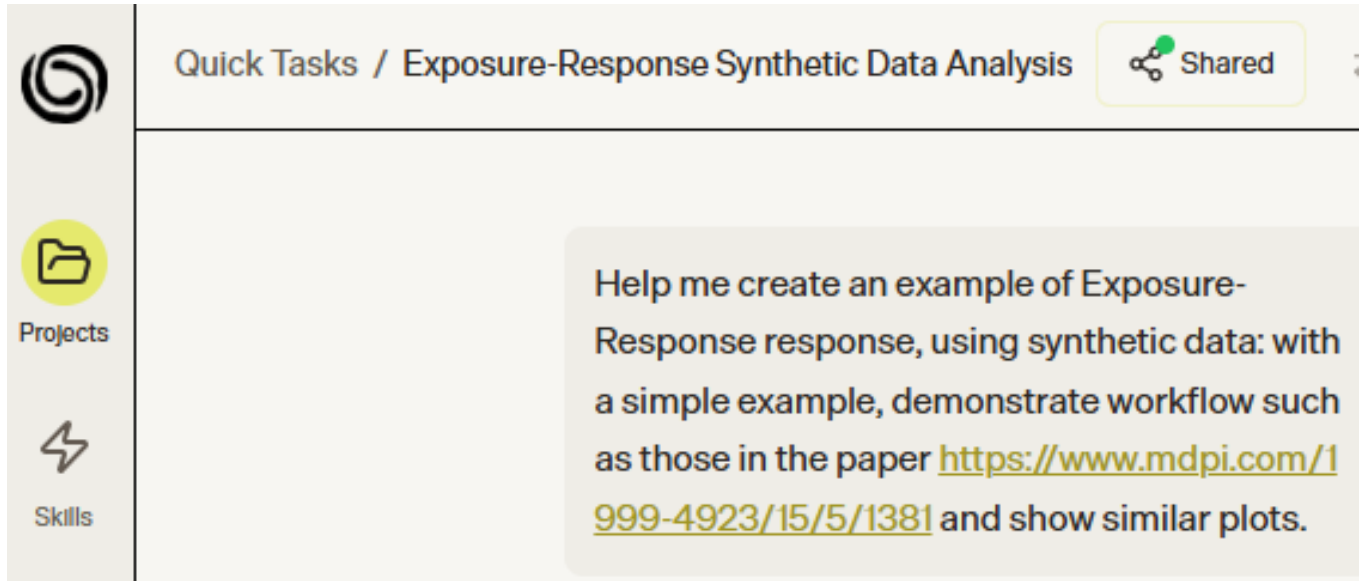


Biomni example: PK/PD





Biomni example: Exposure-Response Analysis



Neural Differential Equations for Dynamic Modeling



From Classical to AI-Partnered Modeling

Classical Modeling



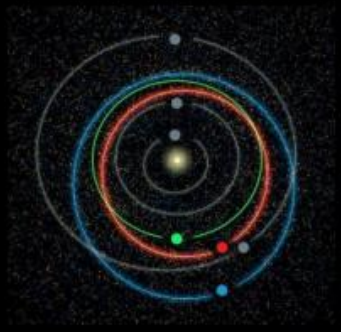


abstraction

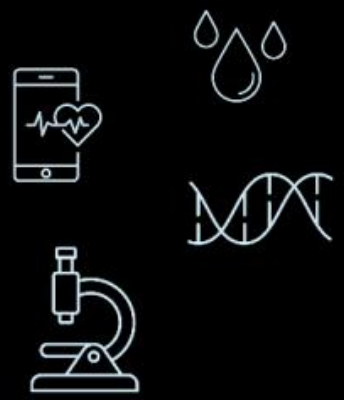


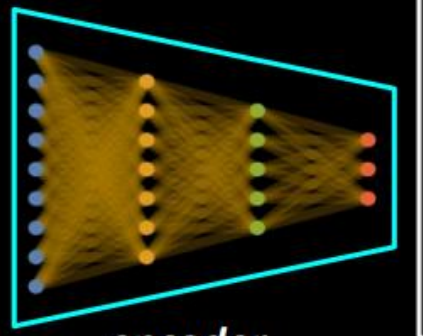


dynamical law

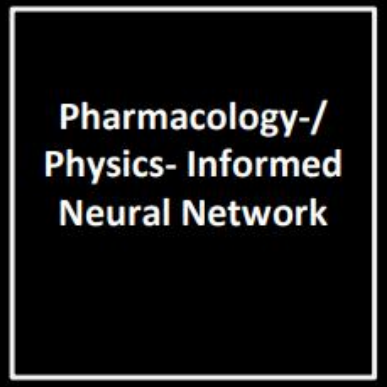


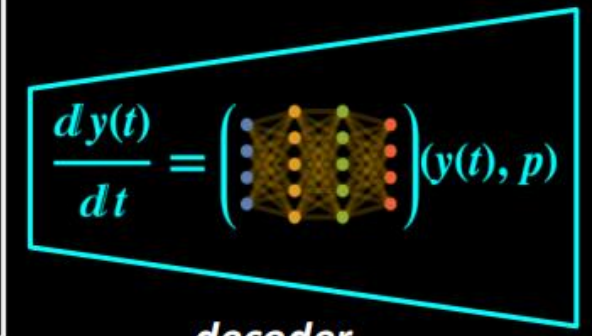
AI-Partnered Modeling



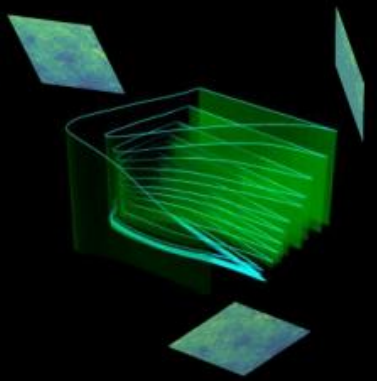


encoder





decoder

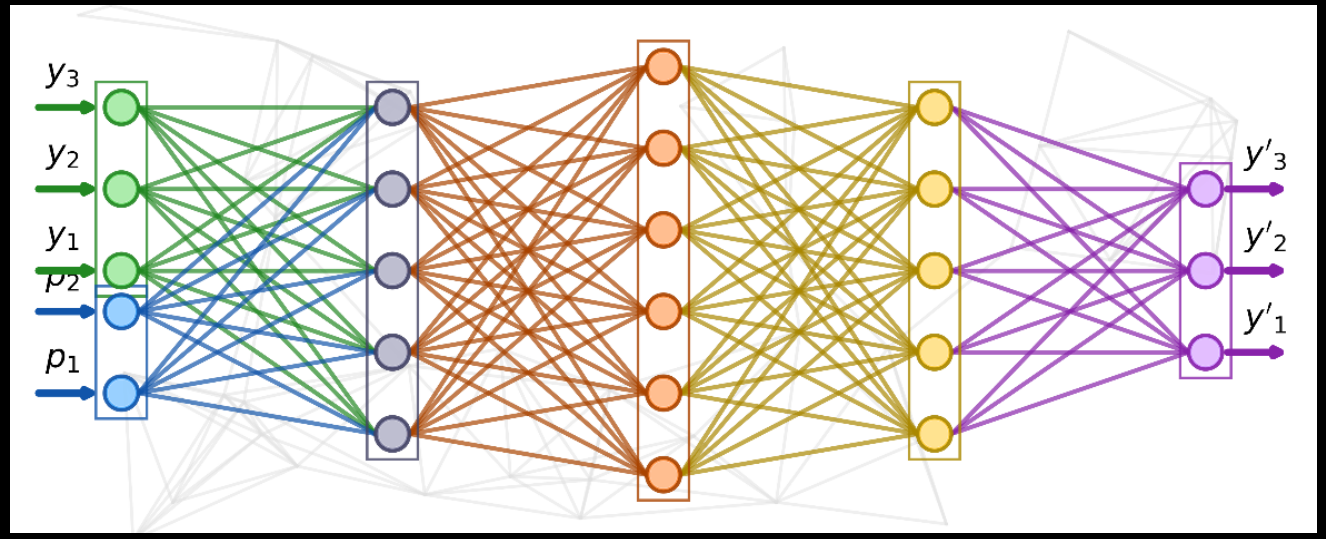


Neural Ordinary Differential Equations

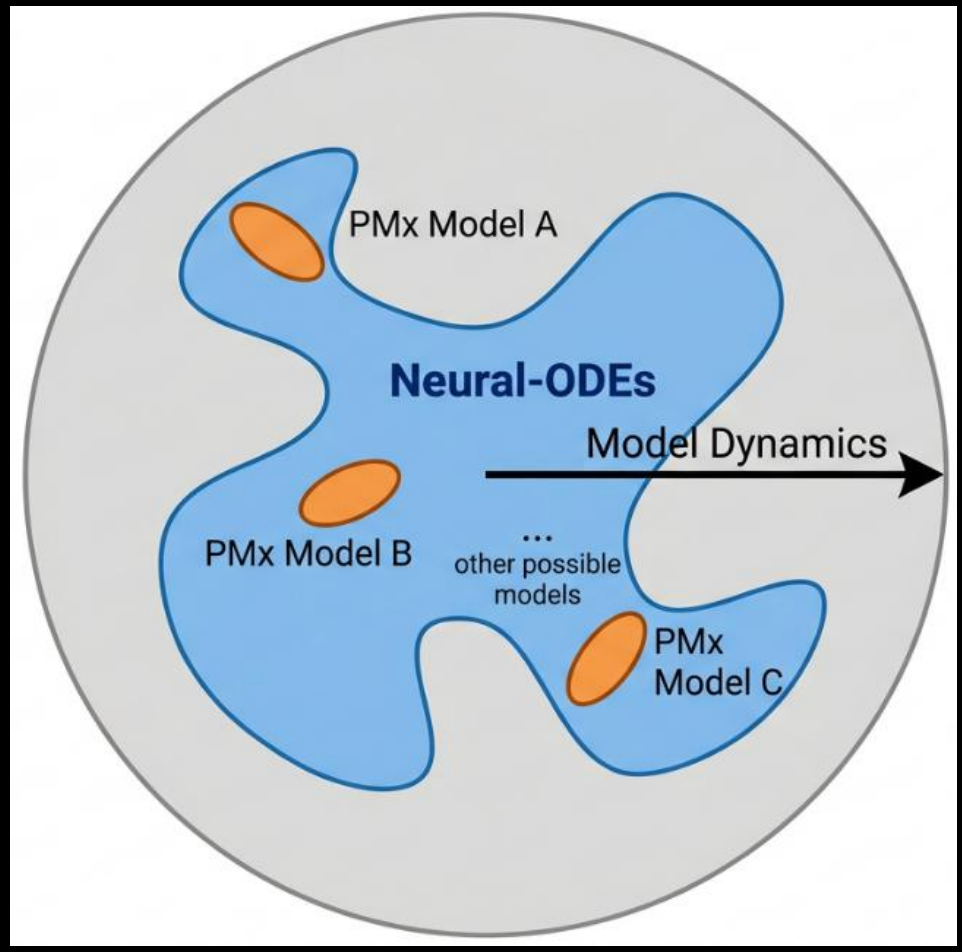
PMx $$\frac{dy(t)}{dt} = f(y(t), p), y(t) \in \mathbb{R}^n, p \in \mathbb{R}^m$$

NODE $$\frac{dy(t)}{dt} = f_\theta(y(t), p), y(t) \in \mathbb{R}^n, p \in \mathbb{R}^m$$

$f_\theta(y(t), p)$:



Space of Model Dynamics



PMx

$$\frac{dy(t)}{dt} = f(y(t), p), y(t) \in \mathbb{R}^n, p \in \mathbb{R}^m$$

NODE

$$\frac{dy(t)}{dt} = f_\theta(y(t), p), y(t) \in \mathbb{R}^n, p \in \mathbb{R}^m$$

Space of Model Dynamics: reproducing Simeoni model

Simeoni Tumor Growth Inhibition (TGI) Model

$$\frac{dx_1(t)}{dt} = \frac{\lambda_0\, x_1(t)}{\left[1+\left(\frac{\lambda_0}{\lambda_1}\, w(t)\right)^{\psi}\right]^{1/\psi}} - k_2\, c(t)\, x_1(t)$$

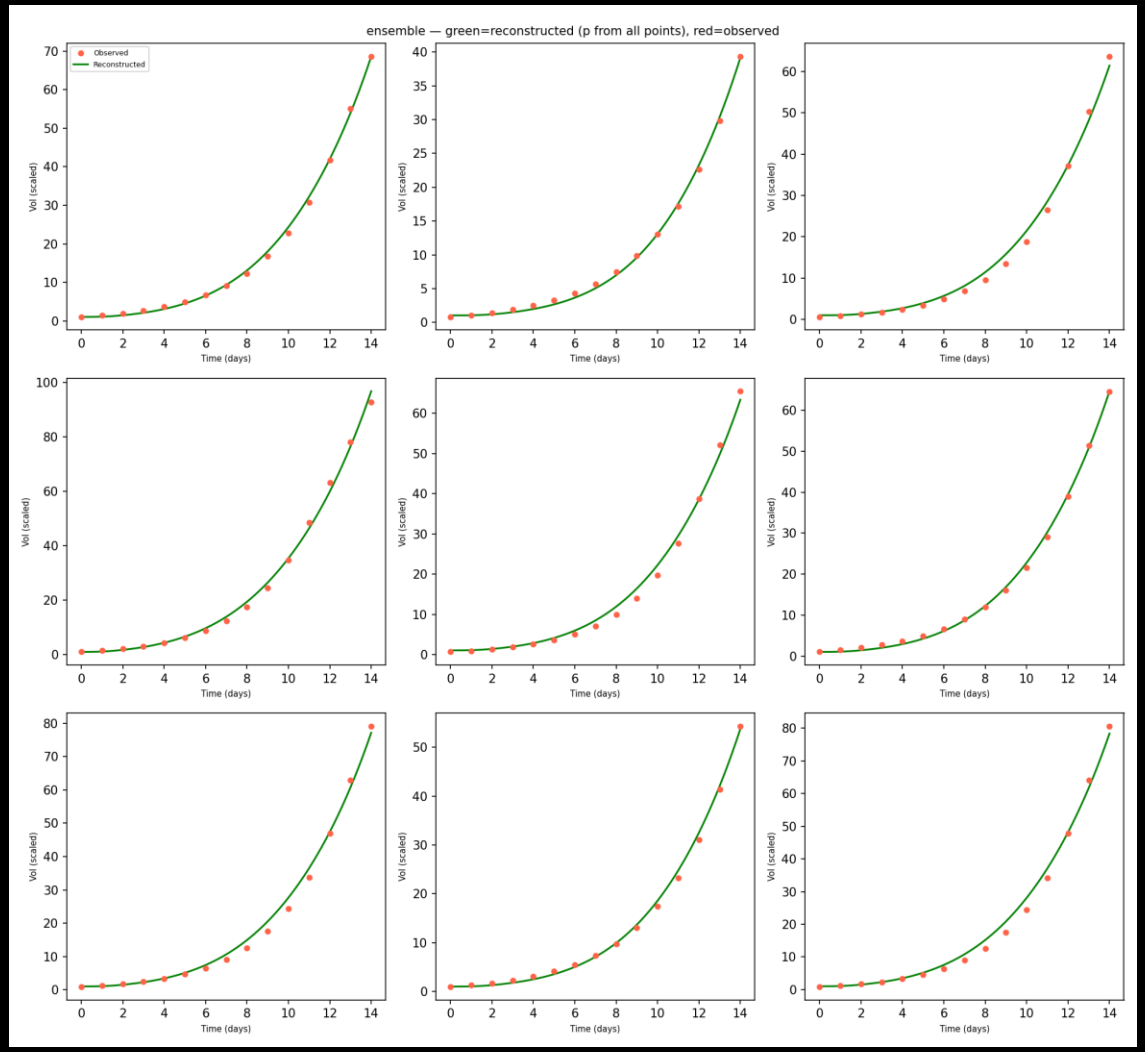
$$\frac{dx_2(t)}{dt} = k_2\, c(t)\, x_1(t) - k_1\, x_2(t)$$

$$\frac{dx_3(t)}{dt} = k_1\, x_2(t) - k_1\, x_3(t)$$

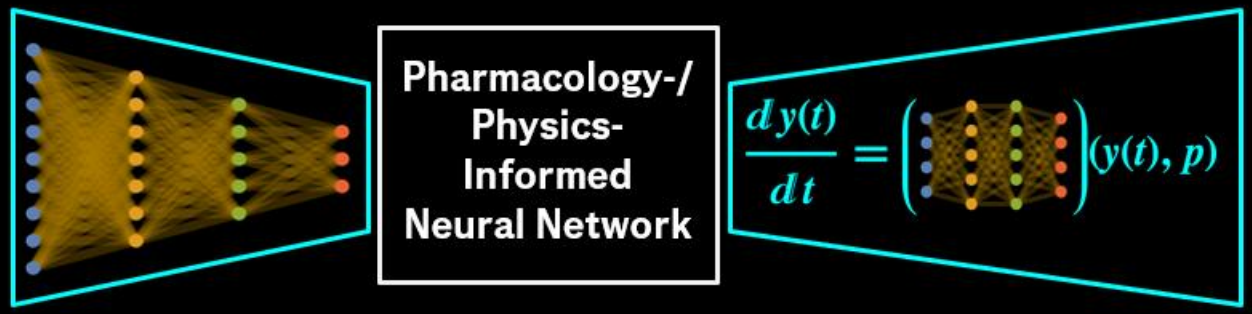
$$\frac{dx_4(t)}{dt} = k_1\, x_3(t) - k_1\, x_4(t)$$

$$w(t) = x_1(t) + x_2(t) + x_3(t) + x_4(t)$$

Initial Conditions ($t = t_0$): $x_1(t_0) = w_0, \quad x_2(t_0) = x_3(t_0) = x_4(t_0) = 0$



Informing Neural Networks with Pharmacology & Physics



Pharmacology Concepts

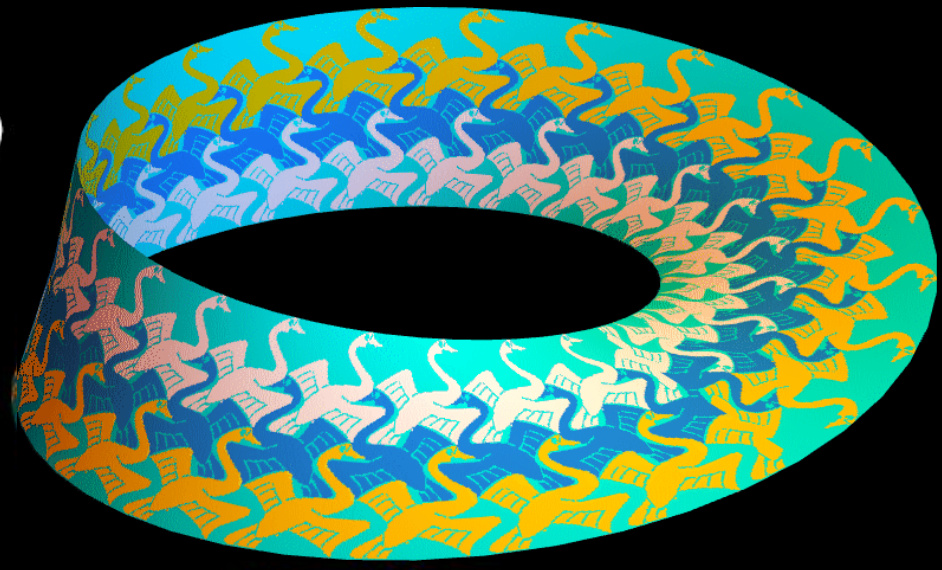
Express *causal* relationships (eg, between *dose*, *PK*, *PD)*
Learn a *dynamical law* from population data
Enable *"what-if"* simulations

Physics Concepts

Imbue vector embeddings with *physical units*

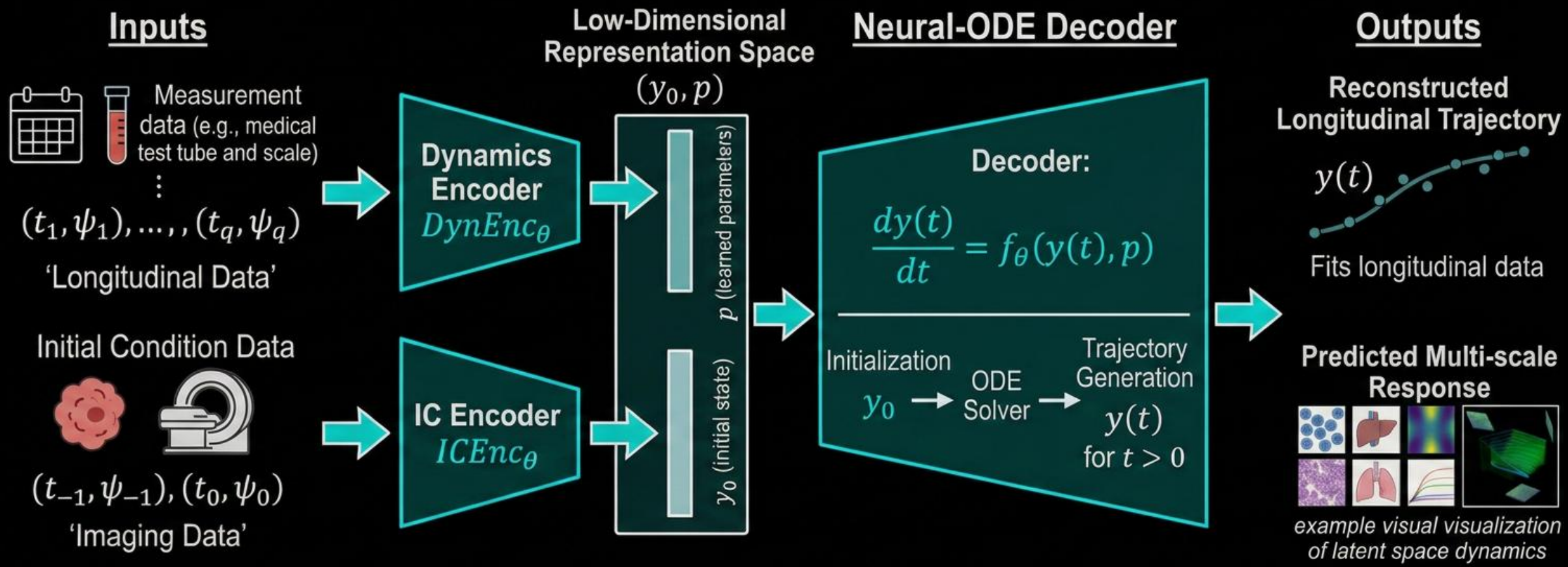
Neural Networks

Learn *representations* of patient data
Improve model as data increases



Lu J *et al. Deep learning prediction of patient response time course from early data via neural-PK/PD modelling*. Nature Machine Intelligence (2021)

Architecture

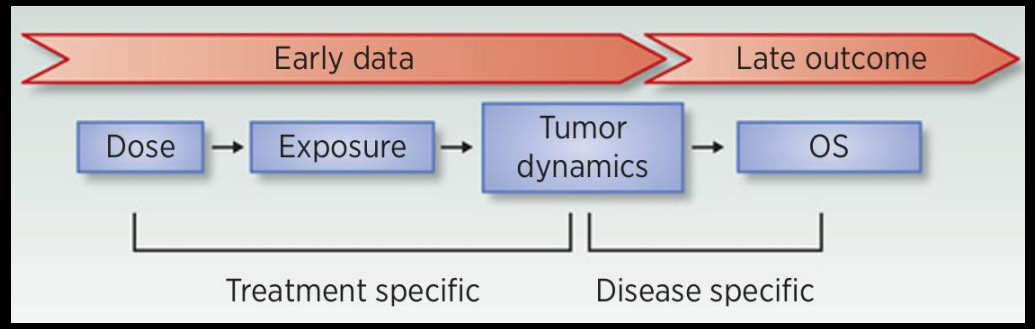


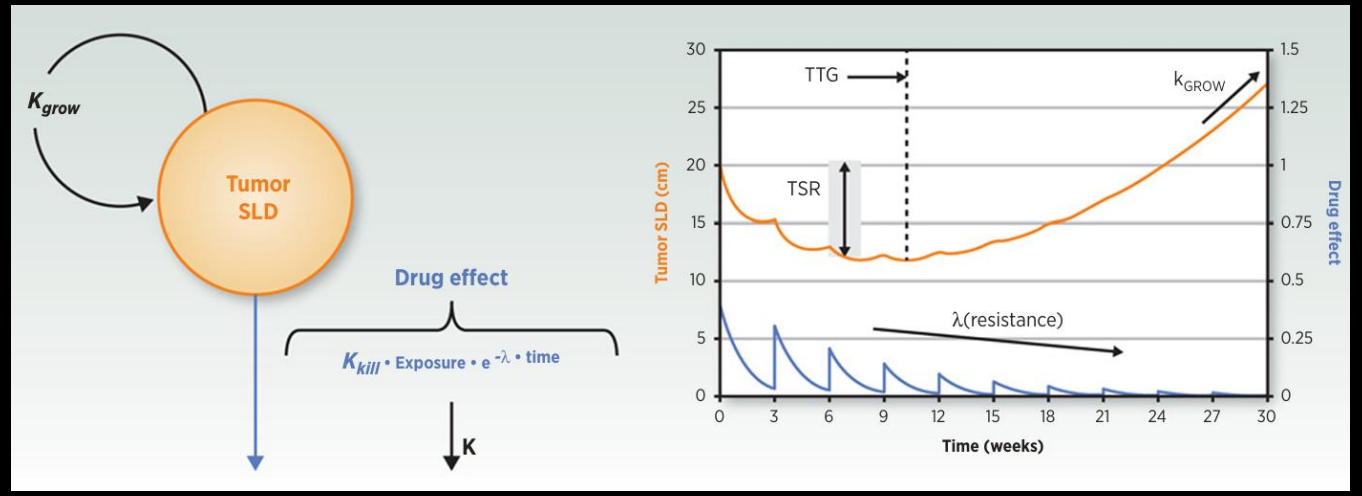
Tumor Dynamic Modeling for Oncology Drug Development

CLINICAL CANCER RESEARCH | REVIEW

Progress and Opportunities to Advance Clinical Cancer Therapeutics Using Tumor Dynamic Models

René Bruno[1], Dean Bottino[2], Dinesh P. de Alwis[3], Antonio T. Fojo[4], Jérémie Guedj[5], Chao Liu[6], Kristin R. Swanson[7], Jenny Zheng[8], Yanan Zheng[9], and Jin Y. Jin[10]



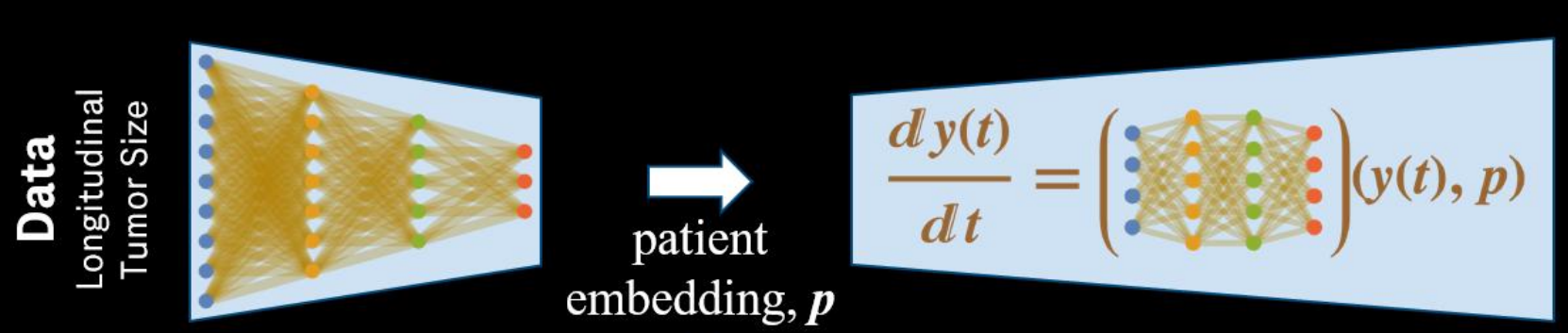


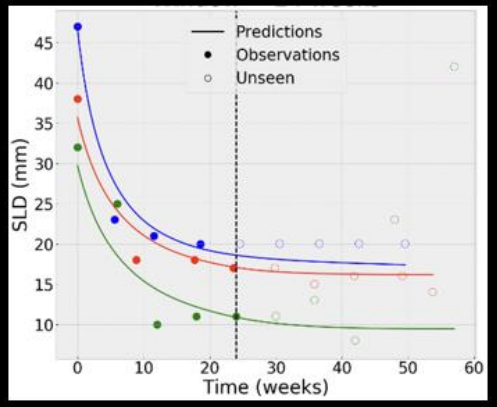
Existing Tumor Dynamic Models

Author	Functional Forms	Treatment
Stein	SLD = BSL · (e^(-kd·t) + e^(kg·t - 1))	Thalidomide, thalidomide + docetaxel, bevacizumab…
Moore	SLD = BSL · e^(-kd·t) + Y1 · (1 - e^(kg·t))	Durvalumab, Durvalumab+ tremelimumab
Claret	dSLD/dt = kg · SLD - kd · SLD kd = kd0 · e^(-λ·t)	Capecitabine; fluorouracil
Wang	SLD = BSL · e^(-kd·t) + kg · t	Paclitaxel and carboplatin; paclitaxel, carboplatin, and bevacizumab; pemetrexed; vinorelbine and cisplatin.
Bonate	SLD = BSL · e^(-kd·t) + kg · t + h · t^2	Pazopanib

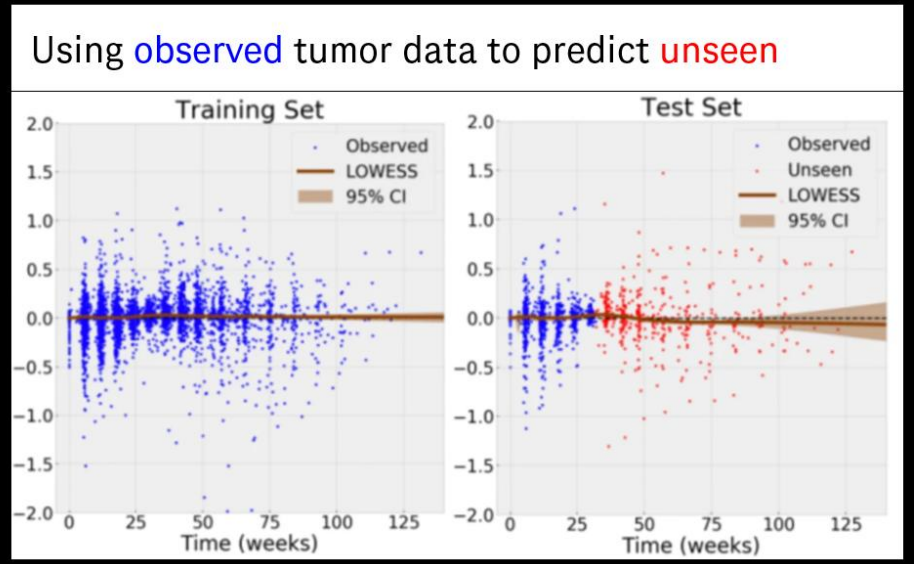
Turner et al, “Follow-up Bias in Tumor Dynamic Modeling: A Comparison of Classical and Neural-ODE Approaches”, CPT:PSP (2026

Application to Tumor Dynamic Modeling



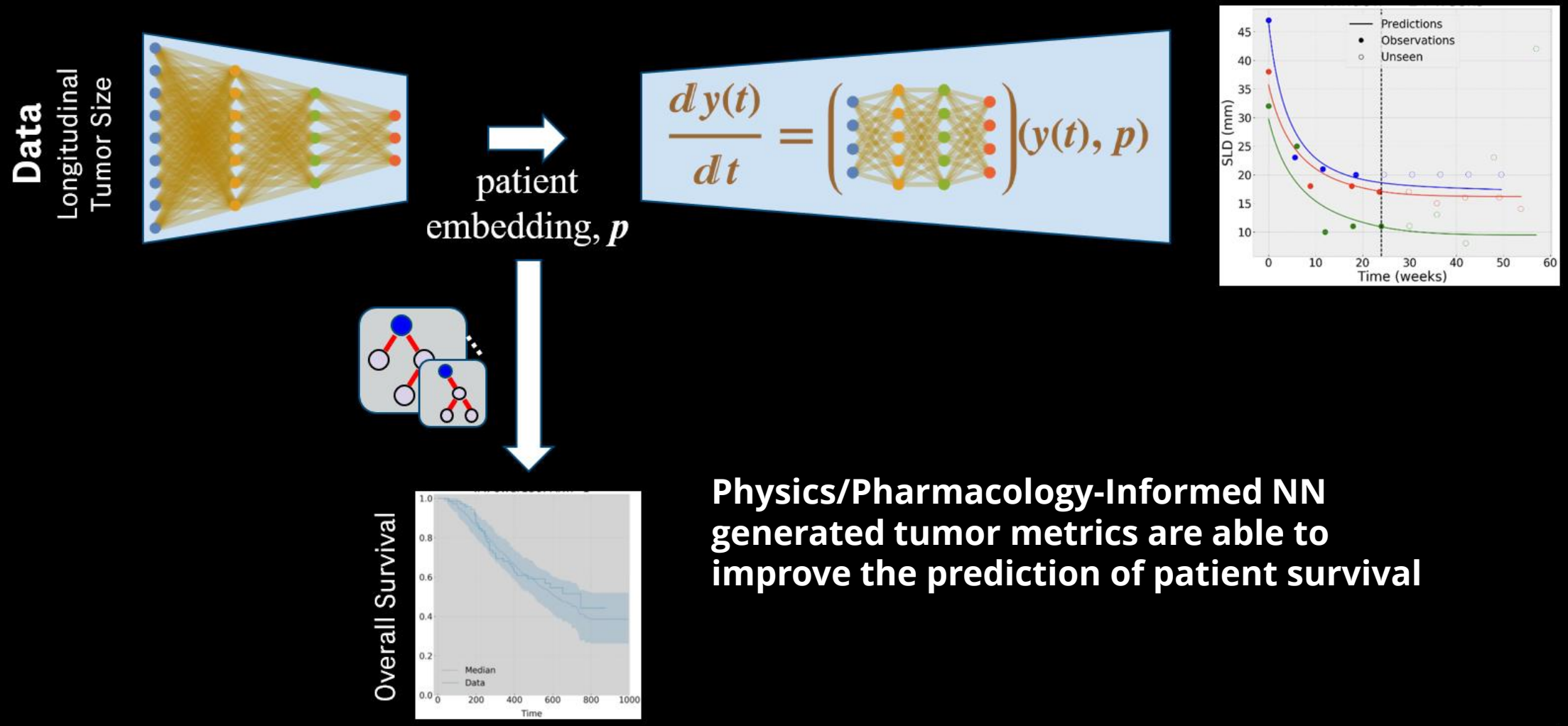


Physics/Pharmacology-Informed Neural-ODE architecture able to predict future tumor sizes from early data



Laurie M, Lu J. npj Systems Biology and Applications. 2023

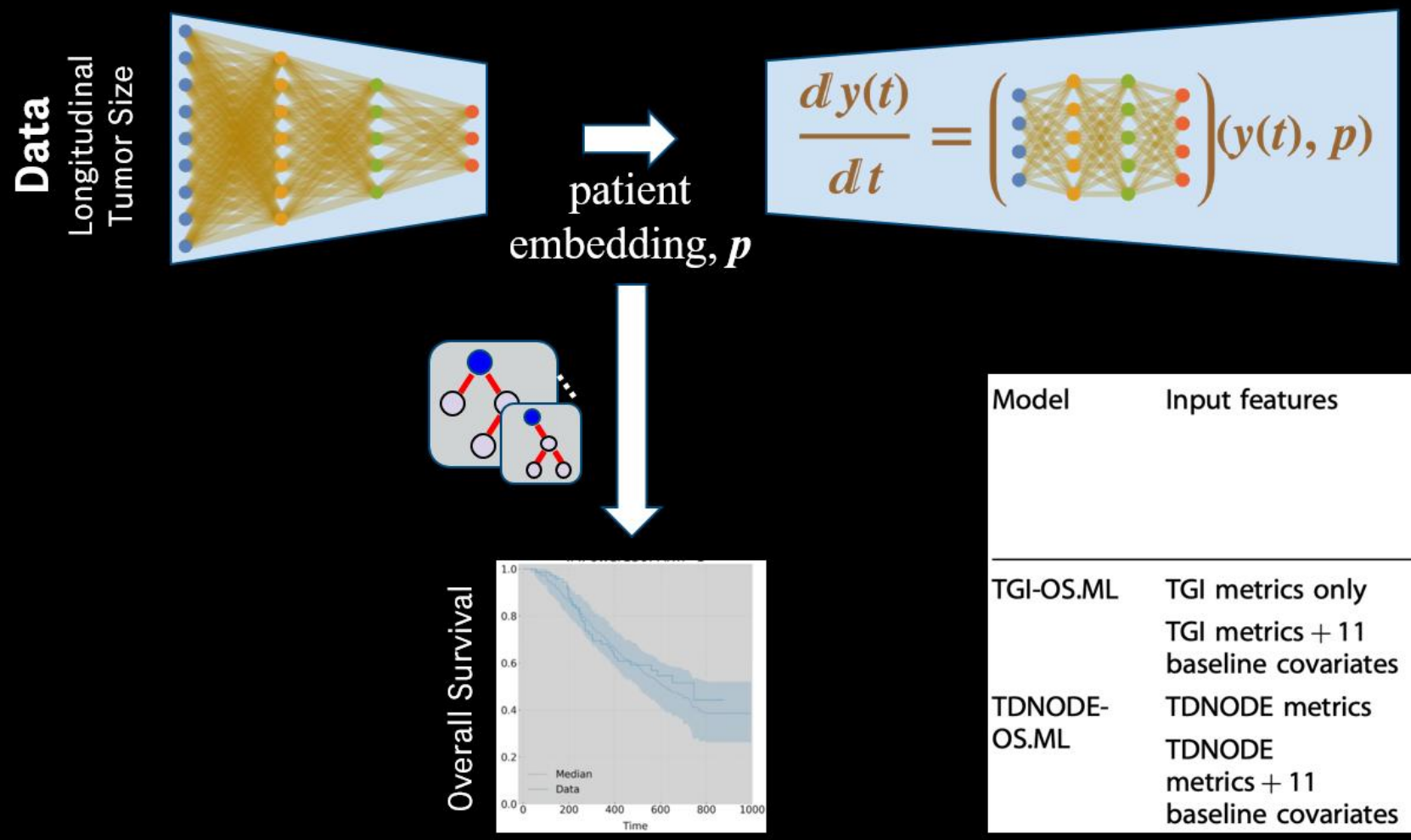
Application to Tumor Dynamic Modeling

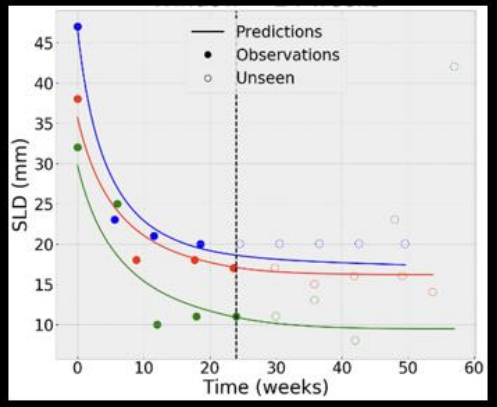


Physics/Pharmacology-Informed NN generated tumor metrics are able to improve the prediction of patient survival

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Application to Tumor Dynamic Modeling



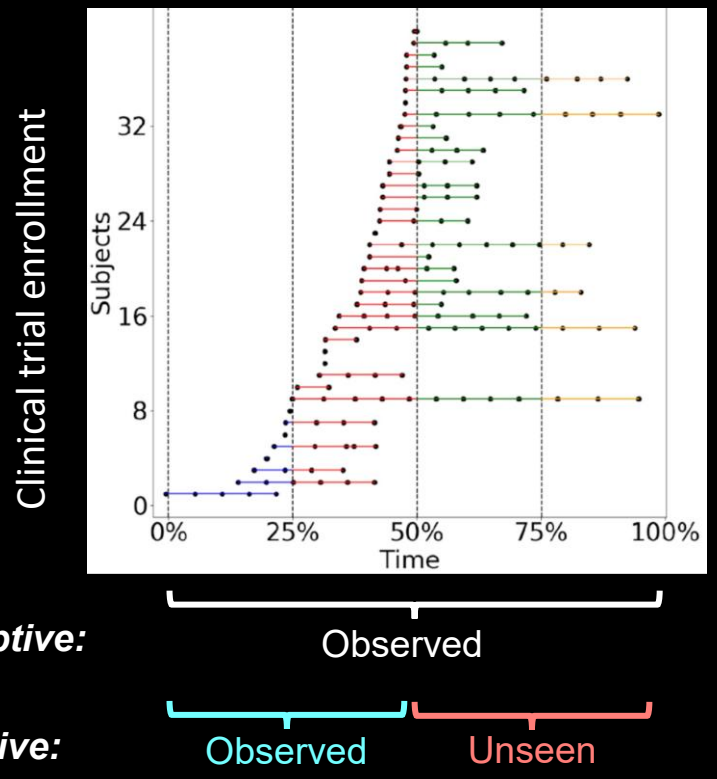


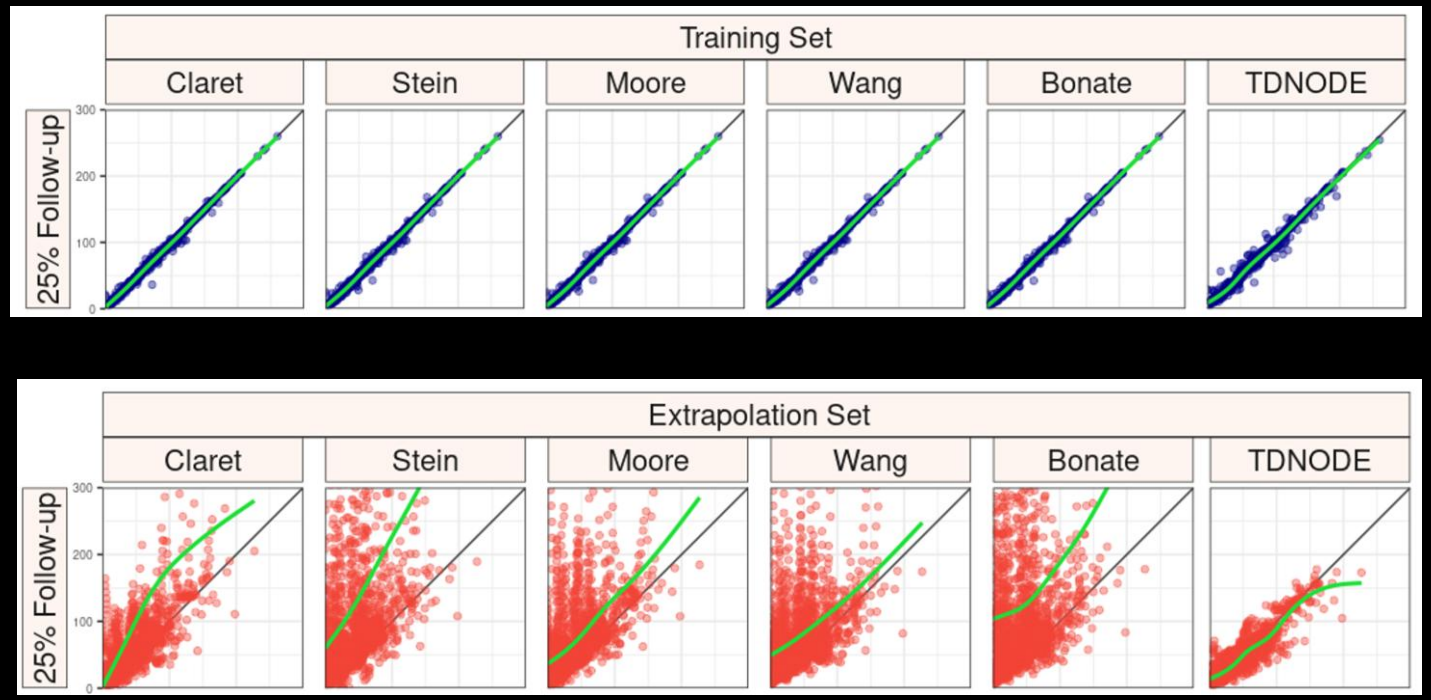
Model	Input features	C-index evaluated via 5-fold cross-validation (median±MAD)	C-index evaluated on the test set
TGI-OS.ML	TGI metrics only	0.73 ± 0.01	0.68
	TGI metrics + 11 baseline covariates	0.78 ± 0.02	0.75
TDNODE-OS.ML	TDNODE metrics	**0.84 ± 0.02**	**0.82**
	TDNODE metrics + 11 baseline covariates	**0.86 ± 0.02**	**0.84**

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Does the Deep Learning Model provide a Step Change?

Numerous Tumor Growth Inhibition (TGI) models already exist for predicting Sum-of-Longest Diameters (**SLD**)
What benefits does TDNODE provide?





Existing TGI models exhibit **bias when estimated using early data**;
TDNODE discovers equations that addresses bias in SLD predictions.

Turner et al, "Follow-up Bias in Tumor Dynamic Modeling: A Comparison of Classical and Neural-ODE Approaches", CPT:PSP (2026).

Does the Deep Learning Model provide a Step Change?

Quantitative comparison of models

TABLE 2 | Quantification of bias and error for TGI and TDNODE models assessed at various values of follow-up times.

	Mean RPE (%)[a,b]	Median RPE (%)[a,b]	R^2[a,c]	RMSE[a,d]
25% enrollment				
Claret	97.11 (26.61)	3.93 (1.42)	0.20 (0.26)	264.47 (221.56)
Stein	334.06 (68.35)	18.32 (3.68)	0.07 (0.03)	141.82 (30.29)
Moore	193.46 (39.60)	9.47 (1.98)	0.34 (0.05)	46.89 (5.00)
Wang	237.81 (41.79)	17.69 (3.43)	0.25 (0.05)	48.22 (4.64)
Bonate	461.75 (93.37)	28.06 (4.74)	0.04 (0.02)	133.76 (31.35)
TDNODE	61.18 (14.72)	−3.09 (1.09)	0.81 (0.02)	16.83 (0.81)
50% enrollment				
Claret	44.64 (5.50)	3.95 (0.59)	0.54 (0.14)	37.31 (10.89)
Stein	137.77 (13.97)	15.71 (1.22)	0.33 (0.04)	57.89 (5.50)
Moore	77.23 (8.19)	6.78 (0.61)	0.75 (0.02)	20.83 (0.93)
Wang	115.55 (10.83)	14.47 (1.22)	0.52 (0.04)	32.97 (1.85)
Bonate	123.25 (11.85)	15.03 (1.17)	0.51 (0.03)	33.61 (1.82)
TDNODE	32.20 (5.15)	−0.77 (0.48)	0.86 (0.01)	14.95 (0.49)

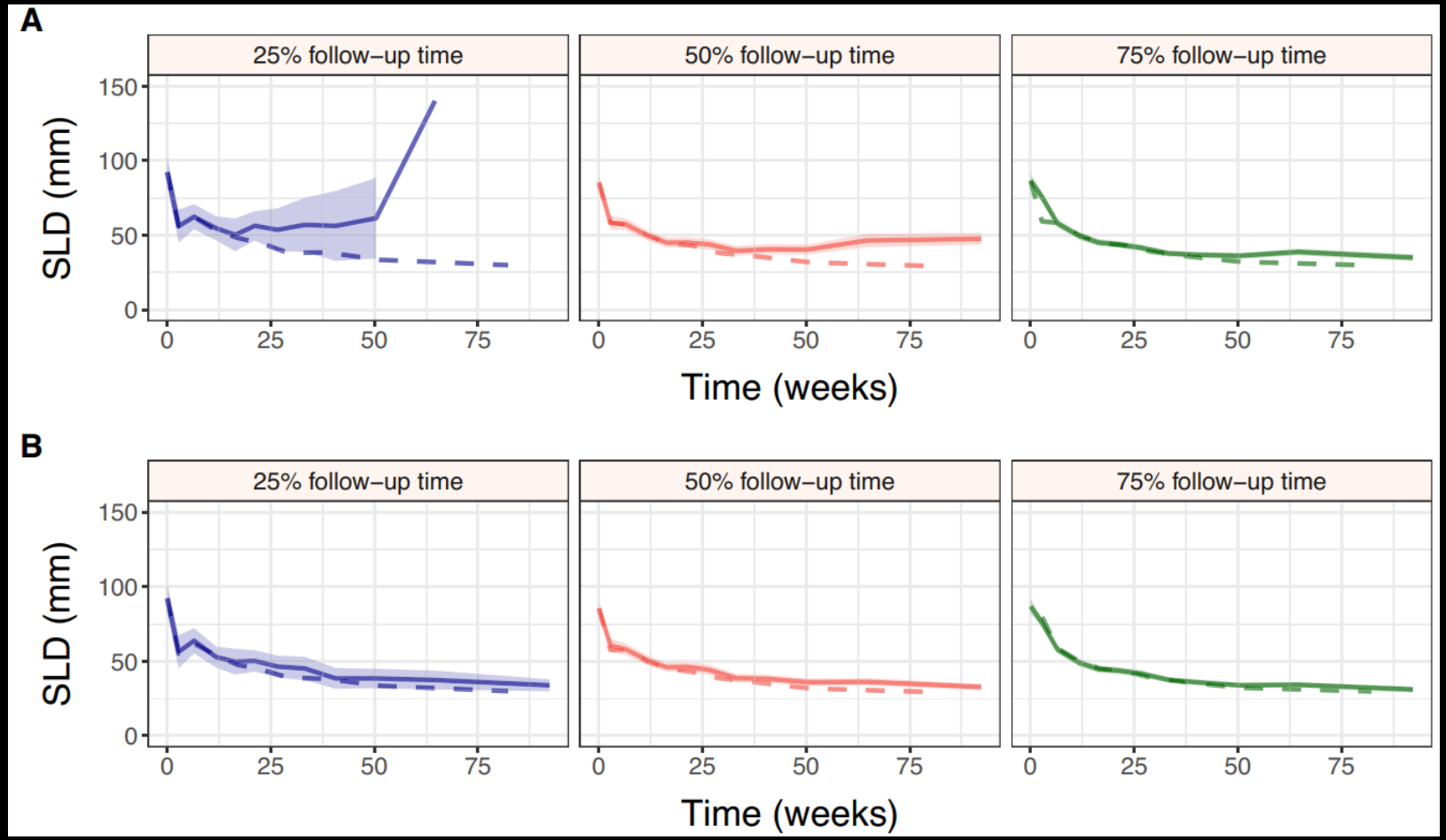
Turner et al, "Follow-up Bias in Tumor Dynamic Modeling: A Comparison of Classical and Neural-ODE Approaches", CPT:PSP (2026).

Does the Deep Learning Model provide a Step Change?

Qualitative comparison of models

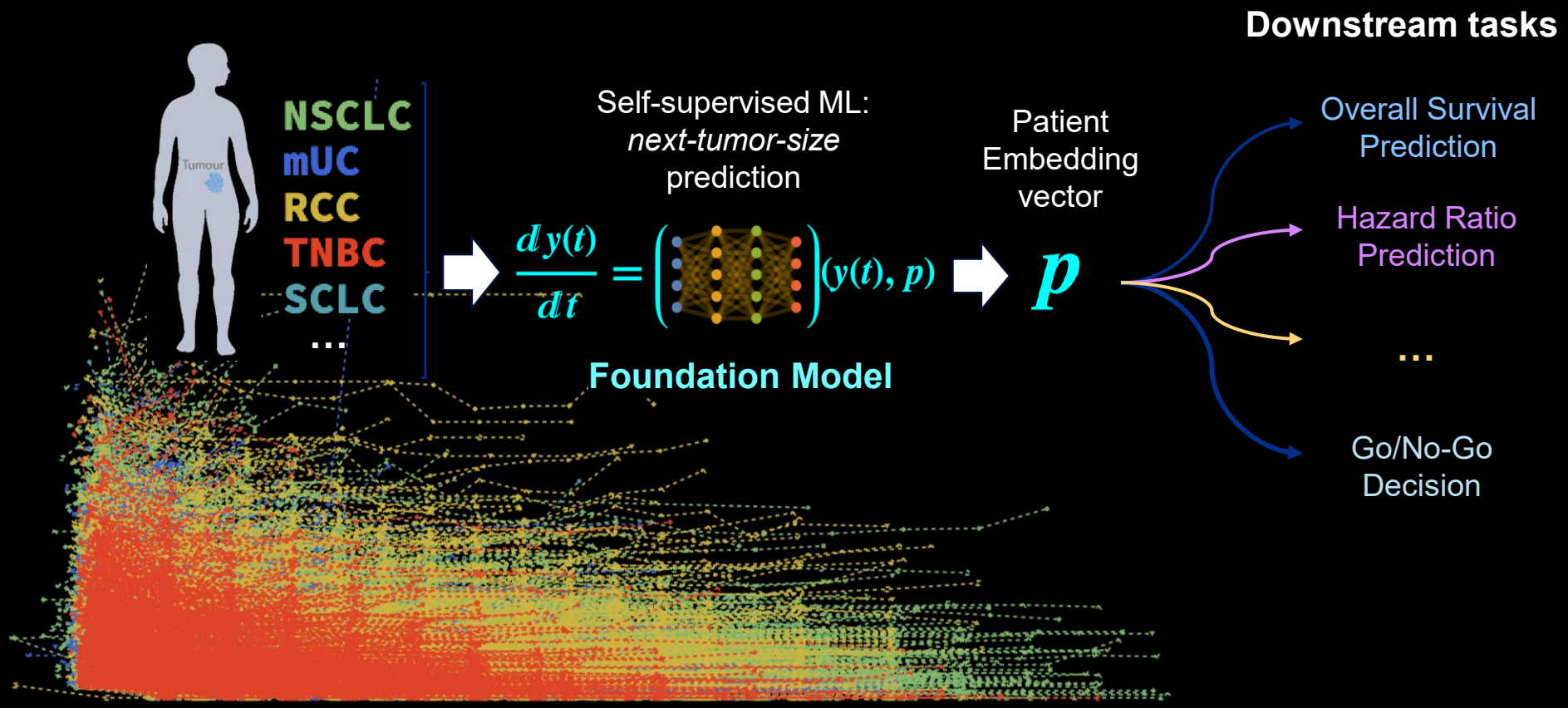
Claret model

TDNODE model

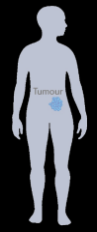


Turner et al, "Follow-up Bias in Tumor Dynamic Modeling: A Comparison of Classical and Neural-ODE Approaches", CPT:PSP (2026).

Towards a Foundation Model for Solid Tumor Dynamics



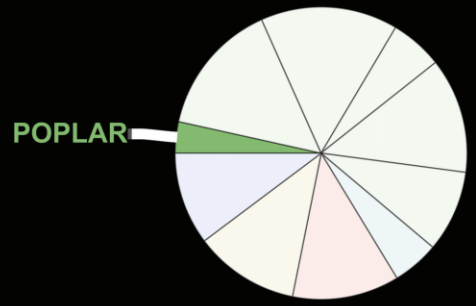
Tumor Dynamics Foundation Model: Predictive across Trials & Tumor Type



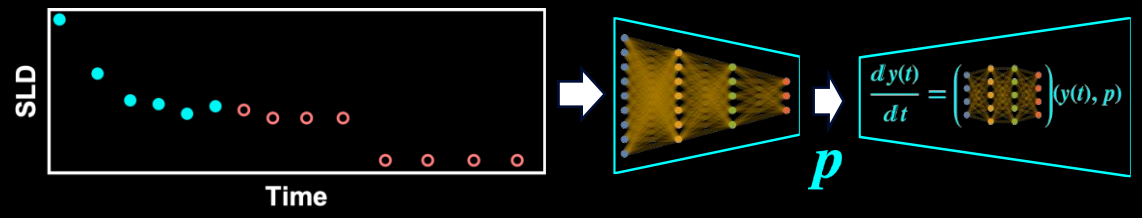
NSCLC
mUC
RCC
TNBC
SCLC

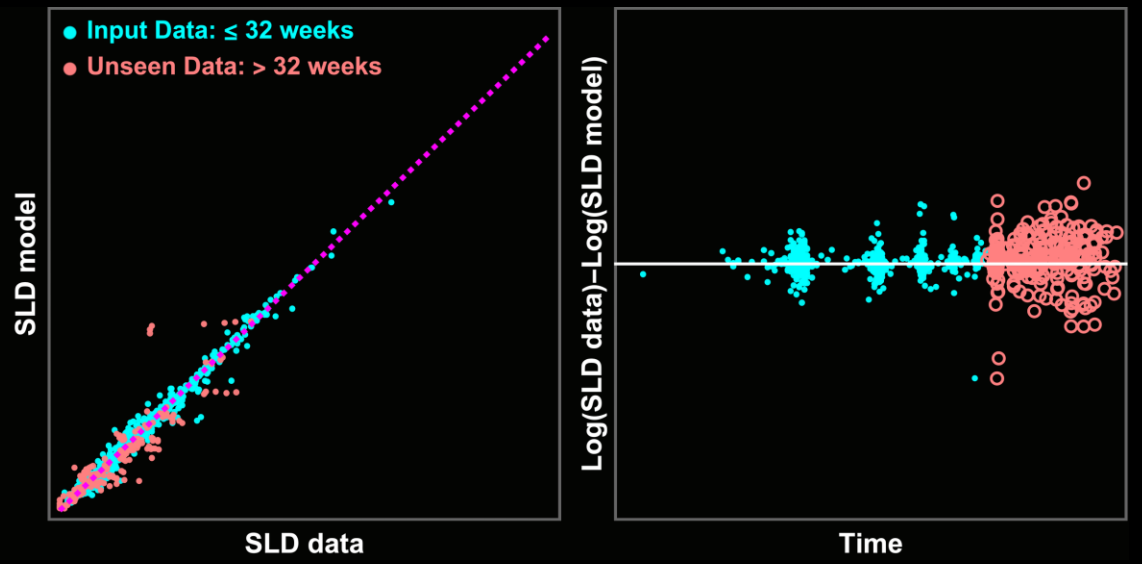
Training set: 9 clinical trials

Test set:

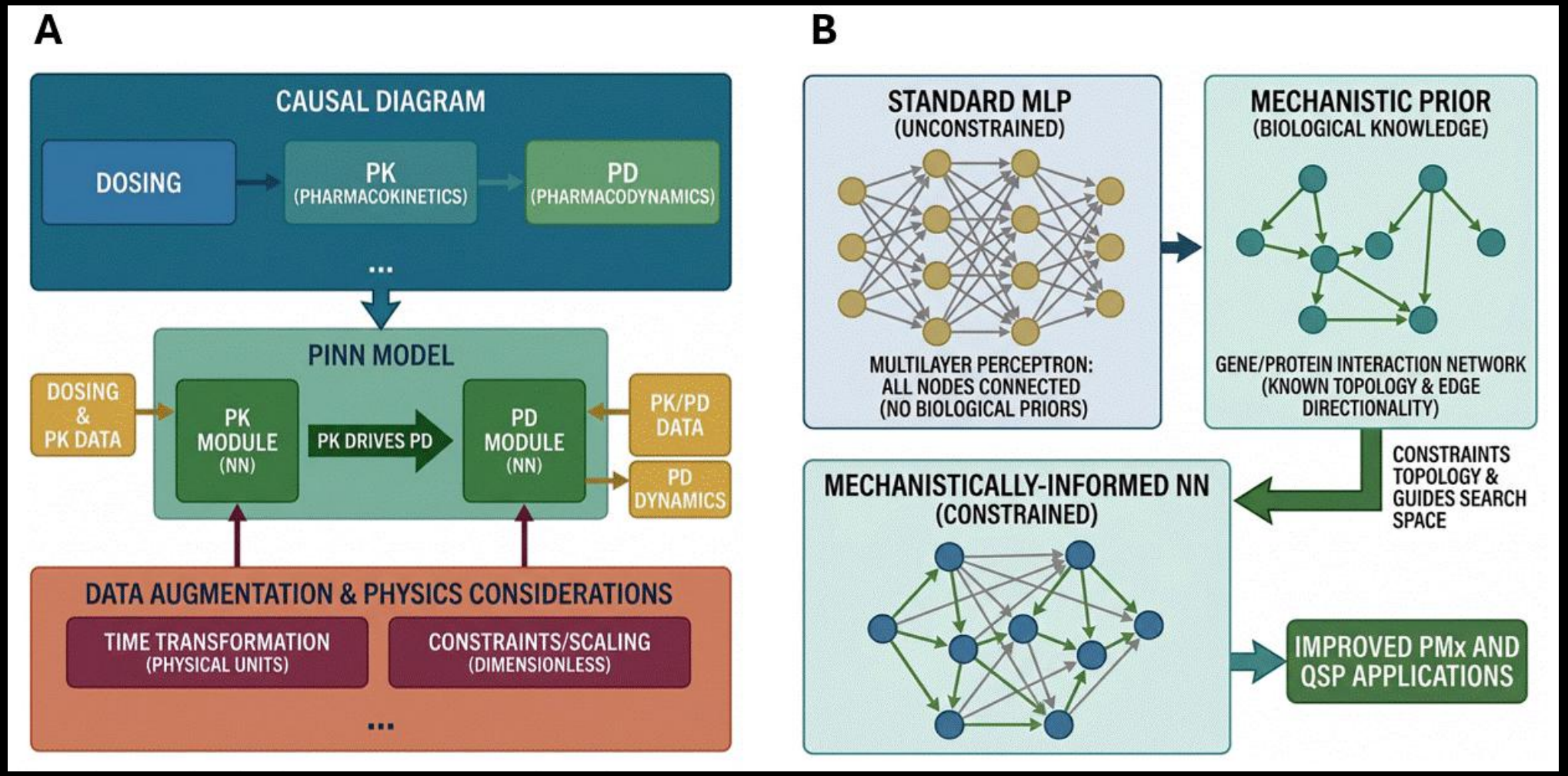


Gengbo Liu





Incorporating Mechanistic Knowledge

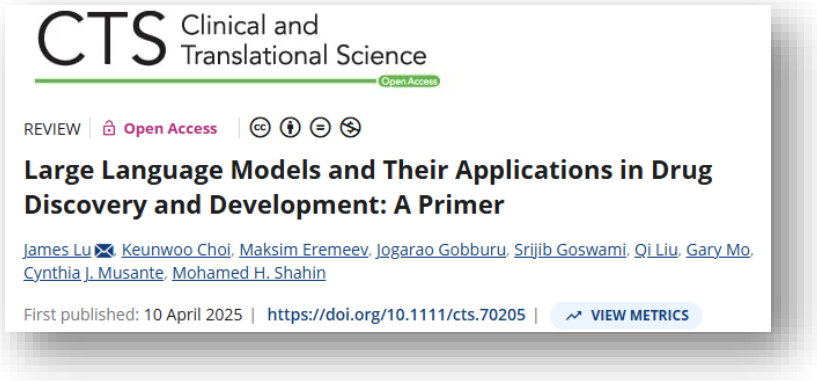


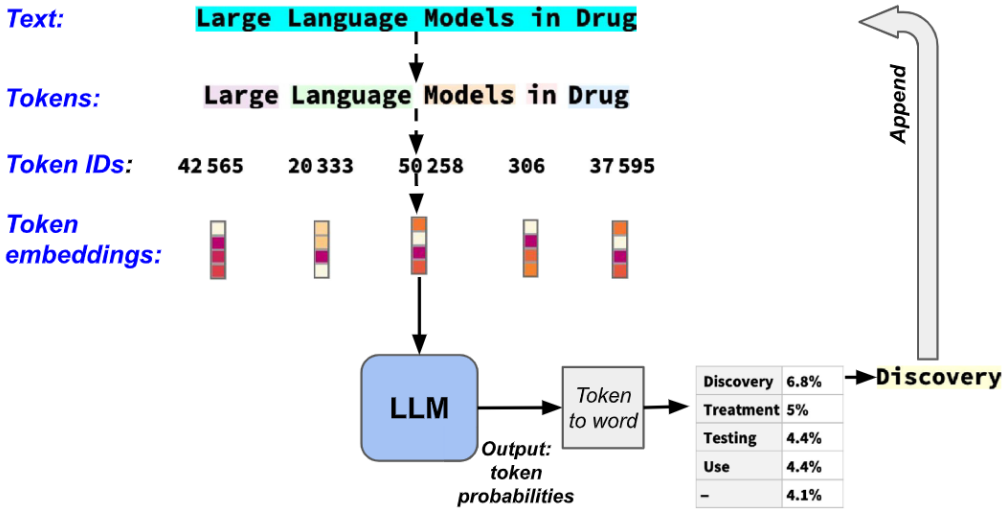
Outlook & Discussions

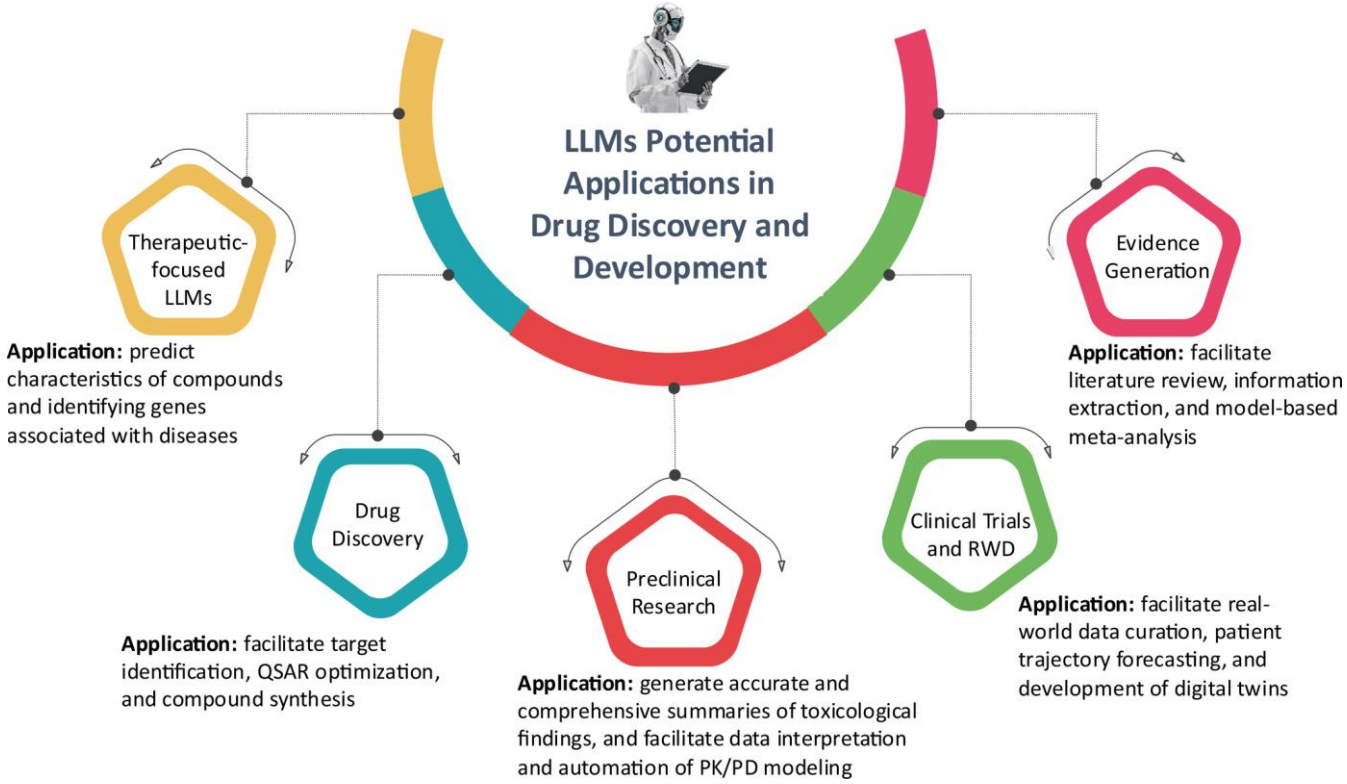


The Rise of Large Language Model

- Transformative potential of AI models with capability to read texts and code offers many opportunities in drug discovery & development





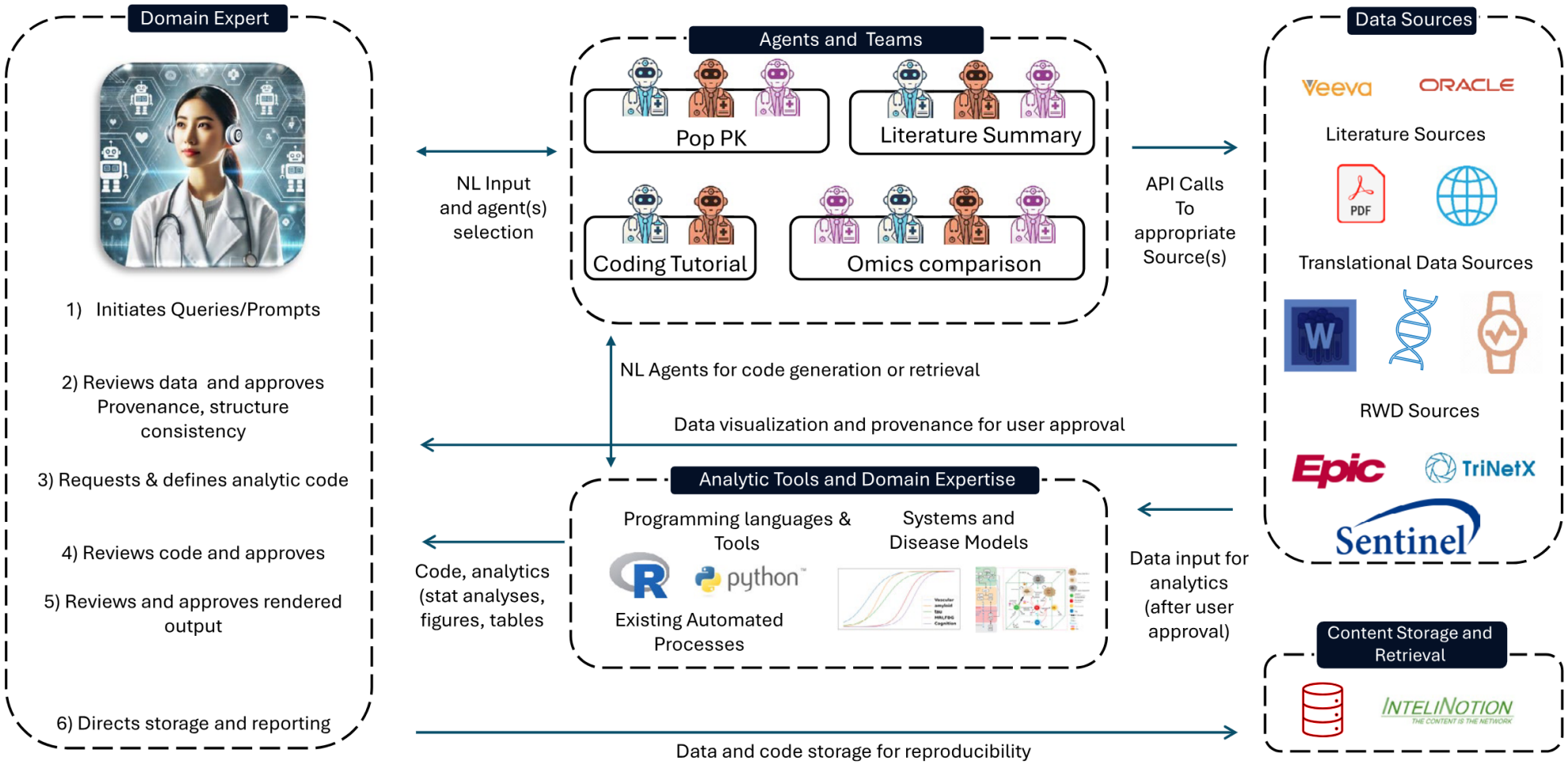


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Agents for Change: Artificial Intelligent Workflows for Quantitative Clinical Pharmacology and Translational Sciences

Mohamed H. Shahin, Srijib Goswami, Sebastian Lobentanzer, Brian W. Corrigan

Agentic Structured Workflow Components in QCP and TCS



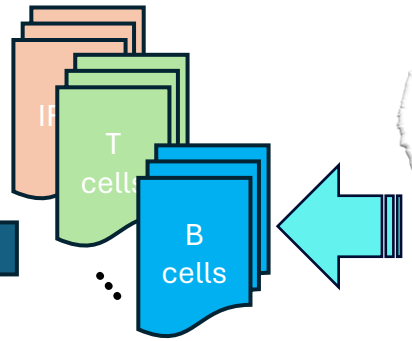
Next-Gen Quantitative Systems Pharmacology Modeling

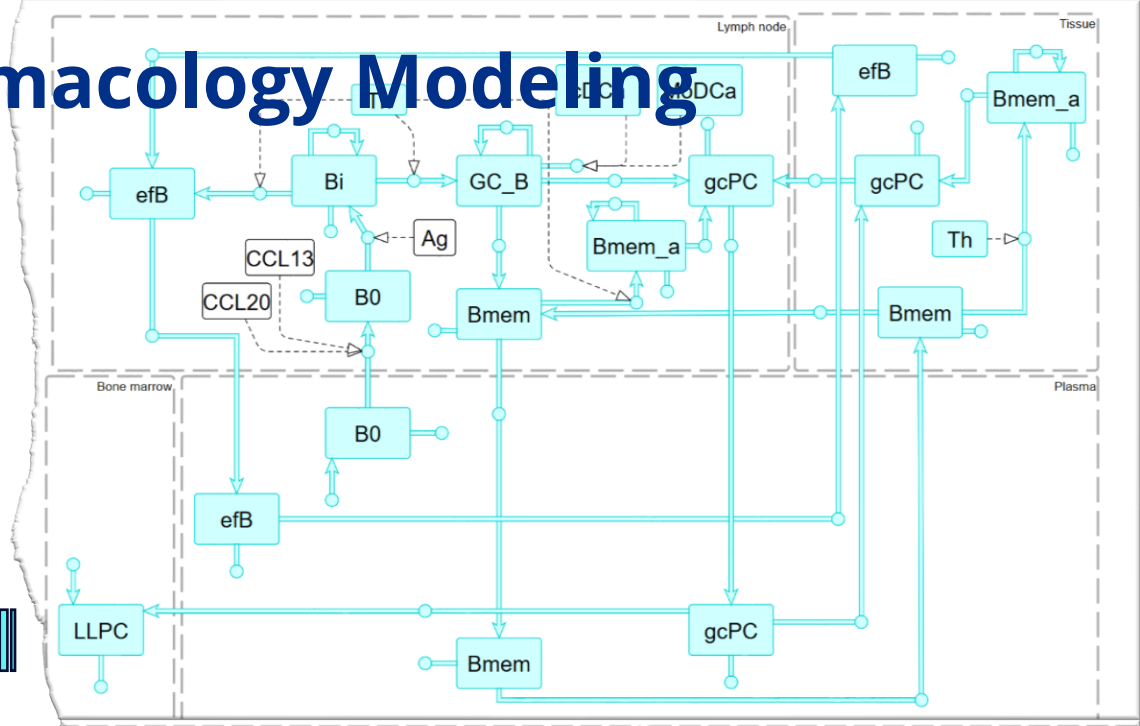
Orchestration of Mathematics & Coding Agents for Multiscale Dynamical Model Discovery

Mathematical Model Equations

$$\frac{dx}{dt} = \alpha \times x - Th0 \ldots$$

Model Generation Agent



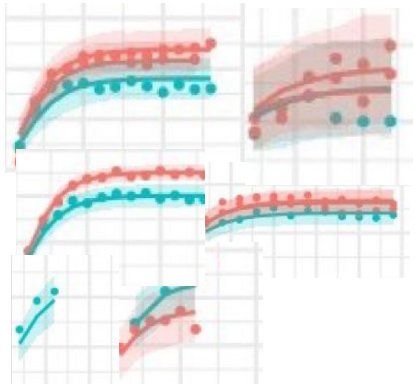


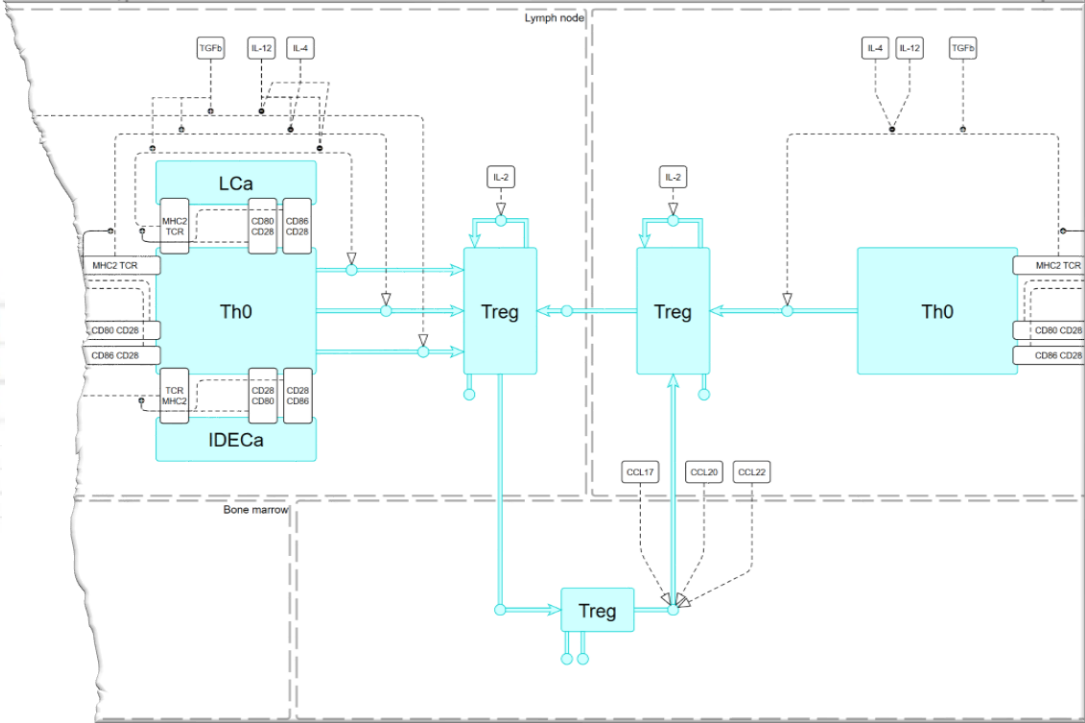
Simulation Dynamics

Scientist-in-the-loop

Calibration Agent

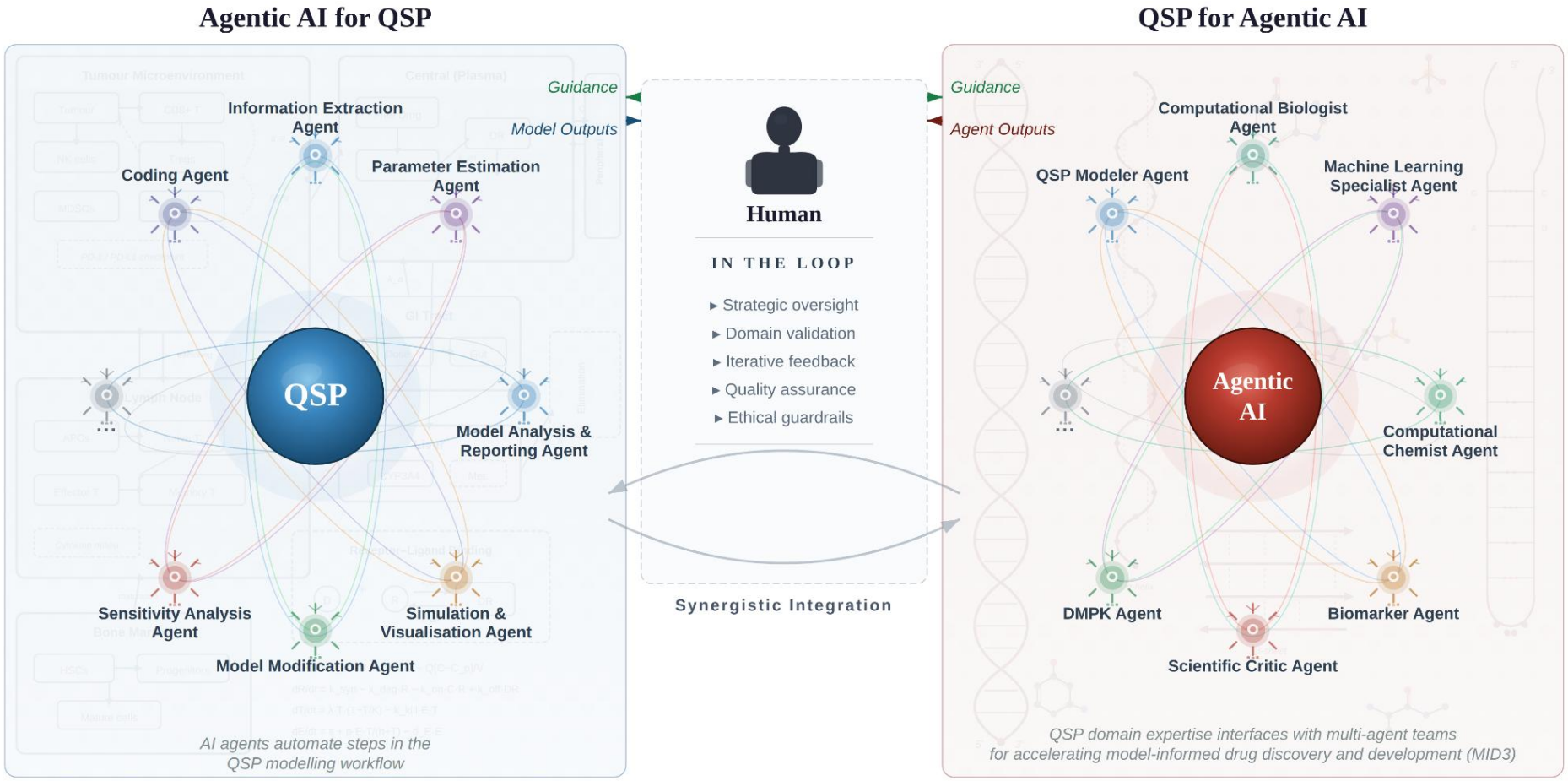
Clinical Data





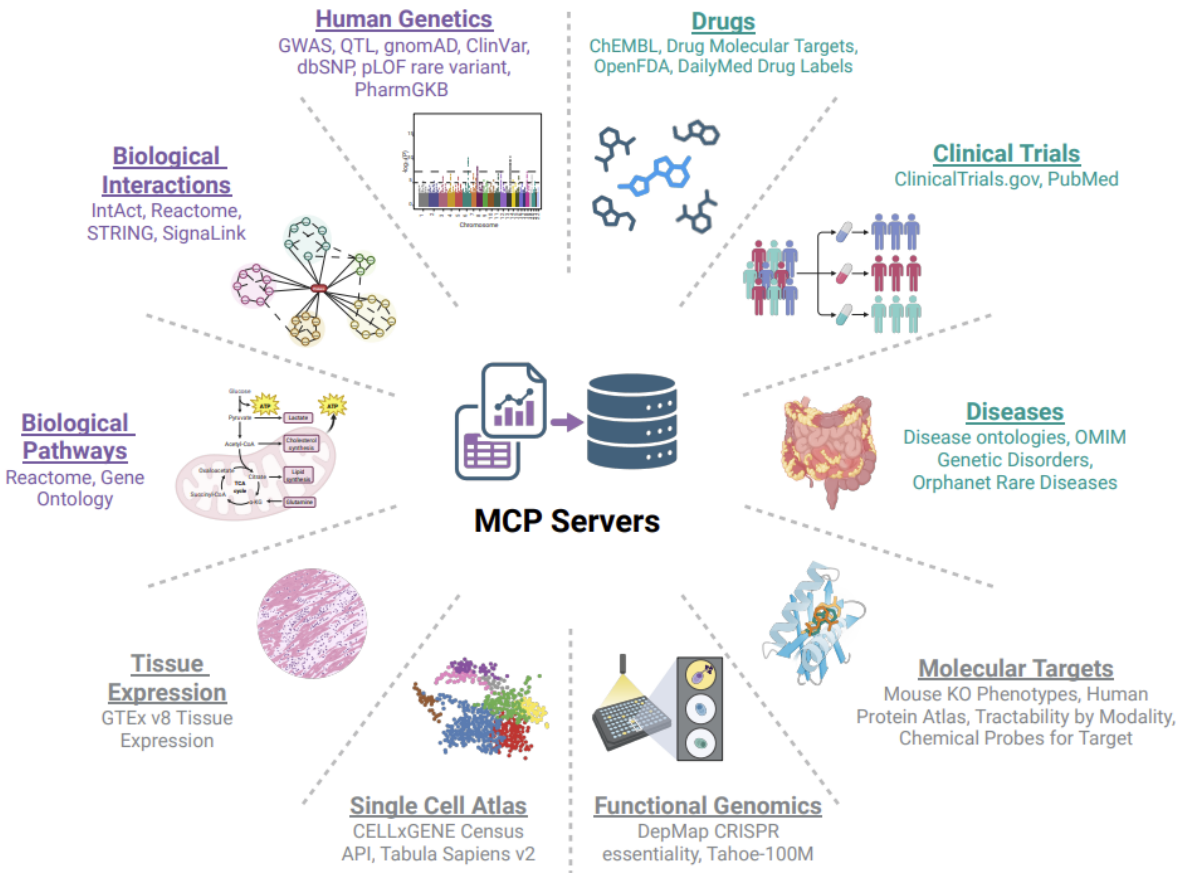
Incorporating Virtual Quantitative Systems Pharmacology (QSP) Modeler into a Multi-Agent Drug Discovery Team

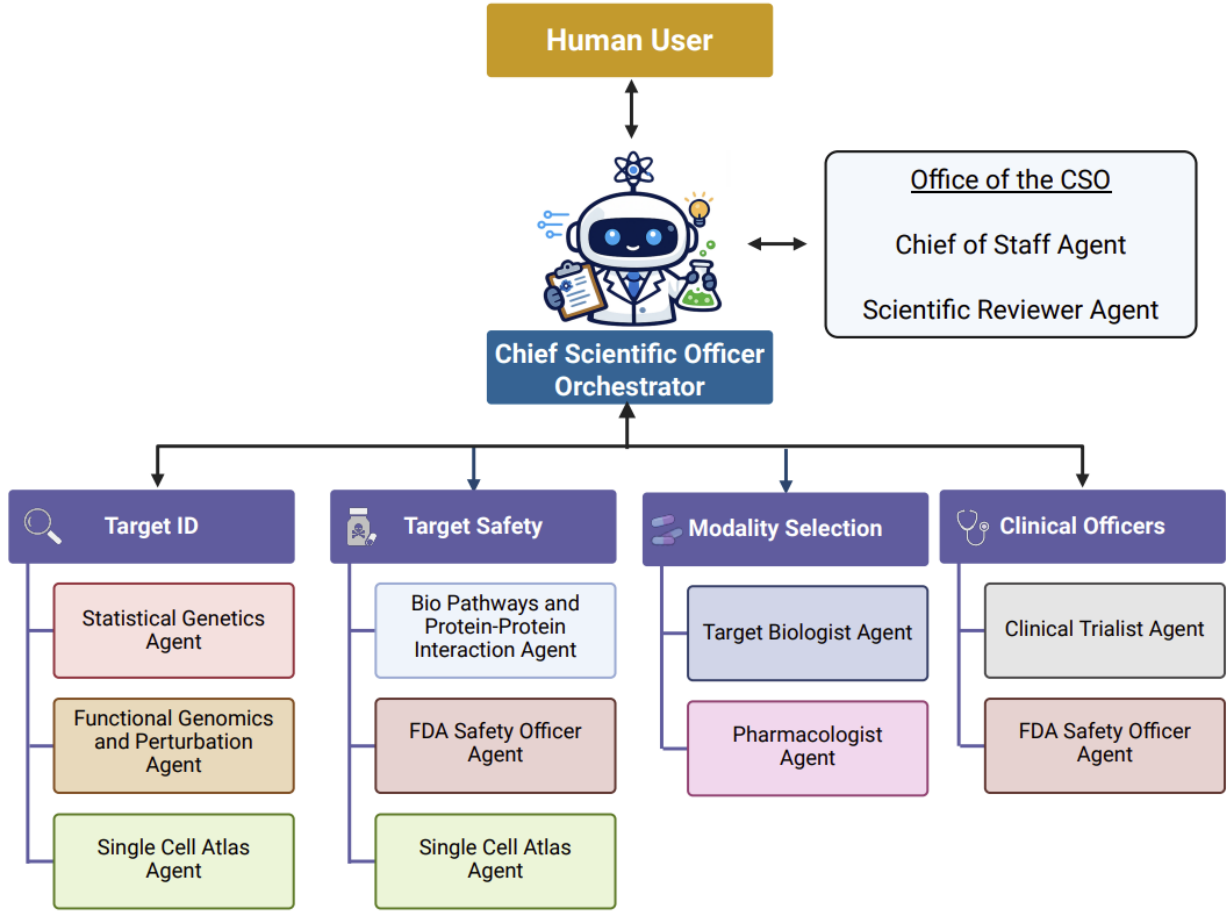
Systems view of diseases and treatments enabled by Hierarchical Multi-Agent System



J. Lu & R. Desikan, "Quantitative Systems Pharmacology Modeling Amid the Rise of Agentic AI", CPT:PSP (2026).

The Future of Drug Discovery Industry?





Zhang, Harrison G., et al. "The Virtual Biotech: A Multi-Agent AI Framework for Therapeutic Discovery and Development." *bioRxiv* (2026)

Concluding Remarks

- In math & coding, AI has advanced significantly – with potential implications for disciplines such as PMx

- Opportunities for the exploitations of AI/ML algorithms underway:
  - “Foundation” / pre-trained models
  - Small *n* but multi-modal

- Multi-Agent Systems (that can code and reason with pharmacological knowledge) poised to re-configure existing drug discovery & development workflows



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THANK YOU

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